

Some Basic Formulas

1. $(a + b)^2 = a^2 + b^2 + 2ab$
2. $(a - b)^2 = a^2 + b^2 - 2ab$
3. $(a + b)^2 = (a - b)^2 + 4ab$
4. $a^2 - b^2 = (a - b)(a + b)$
5. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
6. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
7. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
8. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
9. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
10. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Concept 1:

(i) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

(ii) $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

If $a = b = c$ then

$$a^3 + b^3 + c^3 = 3abc$$

and $a^2 + b^2 + c^2 - ab - bc - ca = 0$

Example: If $a + b + c = 0$, find $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

Sol. If $a + b + c = 0$

then $a^3 + b^3 + c^3 - 3abc = 0$

$$a^3 + b^3 + c^3 = 3abc$$

$$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3$$

Concept 2:

If $x = \frac{2ab}{a+b}$, then $\frac{x+a}{x-a} + \frac{x+b}{x-b} = 2$ and $x = \frac{4ab}{a+b}$, then $\frac{x+2a}{x-2a} + \frac{x+2b}{x-2b} = 2$

Example: If $x = \frac{2\sqrt{24}}{\sqrt{8} + \sqrt{3}}$, find $\frac{x + \sqrt{3}}{x - \sqrt{3}} + \frac{x + \sqrt{8}}{x - \sqrt{8}} = ?$

Sol. $x = \frac{2\sqrt{3} \cdot \sqrt{8}}{\sqrt{8} + \sqrt{3}}, \frac{x}{\sqrt{3}} = \frac{2\sqrt{8}}{\sqrt{8} + \sqrt{3}},$

By componendo and dividendo

$$\frac{x + \sqrt{3}}{x - \sqrt{3}} = \frac{2\sqrt{8} + \sqrt{8} + \sqrt{3}}{2\sqrt{8} - \sqrt{8} - \sqrt{3}}$$

$$\frac{x + \sqrt{3}}{x - \sqrt{3}} = \frac{3\sqrt{8} + \sqrt{3}}{\sqrt{8} - \sqrt{3}} \quad \dots(i)$$

Similarly,

$$\frac{x}{\sqrt{8}} = \frac{2\sqrt{3}}{\sqrt{8} + \sqrt{3}}, \quad \frac{x + \sqrt{8}}{x - \sqrt{8}} = \frac{2\sqrt{3} + \sqrt{8} + \sqrt{3}}{2\sqrt{3} - \sqrt{8} - \sqrt{3}}$$

$$\frac{x + \sqrt{8}}{x - \sqrt{8}} = \frac{3\sqrt{3} + \sqrt{8}}{\sqrt{3} - \sqrt{8}} \quad \dots(ii)$$

On adding (i) & (ii)

$$\frac{x + \sqrt{3}}{x - \sqrt{3}} + \frac{x + \sqrt{8}}{x - \sqrt{8}} = \frac{3\sqrt{8} + \sqrt{3}}{\sqrt{8} - \sqrt{3}} + \frac{3\sqrt{3} + \sqrt{8}}{\sqrt{3} - \sqrt{8}} = \frac{3\sqrt{8} + \sqrt{3} - 3\sqrt{3} - \sqrt{8}}{\sqrt{8} - \sqrt{3}}$$

$$= \frac{2\sqrt{8} - 2\sqrt{3}}{\sqrt{8} - \sqrt{3}} = \frac{2(\sqrt{8} - \sqrt{3})}{(\sqrt{8} - \sqrt{3})} = 2$$

Concept 3:

Some Important formulas

1. If $x + \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 - 2$
2. If $x + \frac{1}{x} = a$, then $x^3 + \frac{1}{x^3} = a^3 - 3a$
3. If $x + \frac{1}{x} = a$, then $x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$
4. If $x + \frac{1}{x} = a$,
then $x^5 + \frac{1}{x^5} = \left(x^3 + \frac{1}{x^3}\right)\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right)$
5. If $x + \frac{1}{x} = a$, then $x^6 + \frac{1}{x^6} = a^6 - 6a^4 + 9a^2 - 2$

Example 1: If $x + \frac{1}{x} = 4$ then find the value of $x^3 + \frac{1}{x^3}$.

Sol. $\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$

$$4^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right), \quad 64 = x^3 + \frac{1}{x^3} + 3 \times 4,$$

$$x^3 + \frac{1}{x^3} = 64 - 12, \quad x^3 + \frac{1}{x^3} = 52$$

or, we can directly use the formula

$$x^3 + \frac{1}{x^3} = a^3 - 3a \quad \text{where, } \left(x + \frac{1}{x}\right) = a$$

Example 2: If $x + \frac{1}{x} = 6$ then find the value of $x^4 + \frac{1}{x^4}$?

Sol. By using formula

or,

$$\text{If } x + \frac{1}{x} = a \text{ then, } x^4 + \frac{1}{x^4} = a^4 - 4a^2 + 2$$

$$x + \frac{1}{x} = 6, \quad x^2 + \frac{1}{x^2} = 6^2 - 2 = 34$$

$$x^4 + \frac{1}{x^4} = 6^4 - 4(6)^2 + 2 = 1296 - 144 + 2 = 1154$$

$$x^4 + \frac{1}{x^4} = 34^2 - 2 \Rightarrow 1154$$

6. If $x - \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 + 2$
7. If $x - \frac{1}{x} = a$, then $x^3 - \frac{1}{x^3} = a^3 + 3a$
8. If $x - \frac{1}{x} = a$ then $x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right) + 2$
9. If $x - \frac{1}{x} = a$,
then $x^5 - \frac{1}{x^5} = \left(x^3 - \frac{1}{x^3}\right)\left(x^2 + \frac{1}{x^2}\right) - \left(x - \frac{1}{x}\right)$
10. If $x - \frac{1}{x} = a$, then $x^6 + \frac{1}{x^6} = a^6 + 6a^4 + 9a^2 + 2$

Example 3: If $x - \frac{1}{x} = 3$ then find the value of $x^3 - \frac{1}{x^3}$?

Sol. $\left(x - \frac{1}{x}\right)^3 = x^3 - \frac{1}{x^3} - 3x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right)$ or, $x^3 - \frac{1}{x^3} = 3^3 + 3 \times 3 = 36$

$$3^3 = x^3 - \frac{1}{x^3} - 3(3), \quad x^3 - \frac{1}{x^3} = 3^3 + 9 = 27 + 9, \quad x^3 - \frac{1}{x^3} = 36$$

Example 4: If $x - \frac{1}{x} = 7$ then find the value of $x^5 - \frac{1}{x^5}$?

Sol. By using formula

$$x^5 - \frac{1}{x^5} = \left(x^3 - \frac{1}{x^3}\right) \left(x^2 + \frac{1}{x^2}\right) - \left(x - \frac{1}{x}\right) = (7^3 + 3 \times 7)(7^2 + 2) - 7 = 18557$$

Concept 4:

- | | |
|--|---|
| 1. If $x + \frac{1}{x} = 2$, then $x = 1$ | 2. If $x + \frac{1}{x} = -2$, then $x = -1$ |
| 3. If $x + \frac{1}{x} = 1$, then $x^3 = -1$ | 4. If $x + \frac{1}{x} = -1$, then $x^3 = 1$ |
| 5. If $x^2 + \frac{1}{x^2} = 1$, then $x^6 = -1$ | 6. If $x^2 + \frac{1}{x^2} = -1$, then $x^6 = 1$ |
| 7. If $x + \frac{1}{x} = \sqrt{3}$, then $x^3 + \frac{1}{x^3} = 0$, $x^6 = -1$ | |

Example 1: If $a + \frac{1}{a+2} = 0$, find the value of $(a+2)^{10} + \frac{1}{(a+2)^{10}}$?

Sol. $a + \frac{1}{a+2} = 0$

$$(a+2) + \frac{1}{(a+2)} = 0 + 2, \quad (a+2) + \frac{1}{(a+2)} = 2$$

From above equation we get

$$(a+2) = 1$$

$$(a+2)^{10} + \frac{1}{(a+2)^{10}} = (1)^{10} + \frac{1}{(1)^{10}} = 1 + 1 = 2$$

Example 2: If $\left(x + \frac{1}{x}\right)^2 = 3$ then find the value of $x^{24} + x^{18} + x^{12} + x^6 + 1$?

Sol. $x + \frac{1}{x} = \sqrt{3}, \quad x^3 + \frac{1}{x^3} = 3\sqrt{3} - 3\sqrt{3}$

$$x^6 + 1 = 0$$

$$x^6 = -1$$

$$= x^{24} + x^{18} + x^{12} + x^6 + 1$$

$$= (x^6)^4 + (x^6)^3 + (x^6)^2 + x^6 + 1$$

$$= (-1)^4 + (-1)^3 + (-1)^2 + (-1) + 1$$

$$= 1 - 1 + 1 - 1 + 1 = 1$$

Types of Questions

1. If $a + b + c = 0$, then find the value of

$$\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b}?$$

Sol. $\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b}$

$$= \frac{(-c)}{c} + \frac{(-a)}{a} + \frac{(-b)}{b} = -1 - 1 - 1 = -3$$

2. If $a^2 + ab + b^2 = b^2 + bc + c^2$ where $a \neq b \neq c$, then find the value of $a + b + c$.

Sol. $a^2 + ab + b^2 = b^2 + bc + c^2$

$$a^2 + ab = bc + c^2, \quad a^2 - c^2 + ab - bc = 0$$

$$(a - c)(a + c) + b(a - c) = 0, \quad (a - c)(a + c + b) = 0$$

we know that

$$a \neq c$$

$$\text{therefore, } a + b + c = 0$$

3. If $y = 2 - \sqrt{3}$ and $xy = 1$, then find the value of

$$\frac{1}{x^2} + \frac{1}{y^2}?$$

Sol. $xy = 1 \Rightarrow x = \frac{1}{y}$

$$y = 2 - \sqrt{3}, \quad x = \frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}}, \quad x = 2 + \sqrt{3}$$

$$x + y = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$$

$$\frac{1}{x^2} + \frac{1}{y^2} = \frac{x^2 + y^2}{x^2 y^2} = \frac{(x + y)^2 - 2xy}{(xy)^2}$$

$$= \frac{(4)^2 - 2(1)}{1} = 16 - 2 = 14$$

4. If $x + \frac{1}{x} = 1$, then find $x^{85} + x^{83} + x^{80} + x^{88}$?

Sol. $x + \frac{1}{x} = 1$, then $x^3 = -1$

$$= x^{85} + x^{88} + x^{80} + x^{83}$$

$$= x^{85} + x^3 \cdot x^{85} + x^{80} + x^3 \cdot x^{80}$$

$$= x^{85} + (-1)x^{85} + x^{80} + (-1)x^{80}$$

$$= x^{85} - x^{85} + x^{80} - x^{80} = 0$$

5. If $x^2 + \frac{1}{x^2} = 7$, then find the value of $x^3 + \frac{1}{x^3}$?

Sol. $x^2 + \frac{1}{x^2} = 7$

$$x^2 + \frac{1}{x^2} + 2 = 7 + 2, \quad \left(x + \frac{1}{x}\right)^2 = (3)^2, \quad x + \frac{1}{x} = 3$$

taking cube of the above equation.

$$\left(x + \frac{1}{x}\right)^3 = 3^3, \quad x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = 27$$

$$x^3 + \frac{1}{x^3} + 3(3) = 27, \quad x^3 + \frac{1}{x^3} = 27 - 9$$

$$x^3 + \frac{1}{x^3} = 18$$

6. If $x = 3 - 2\sqrt{2}$, then find $x^6 + x^4 + x^2 + 1$?

Sol. $x = 3 - 2\sqrt{2}, \quad \frac{1}{x} = 3 + 2\sqrt{2}$

On adding both the equation.

$$x + \frac{1}{x} = 6, \quad x^3 + \frac{1}{x^3} = 6^3 - 3(6)$$

$$x^3 + \frac{1}{x^3} = 216 - 18, \quad x^3 + \frac{1}{x^3} = 198$$

$$\frac{x^6 + x^4 + x^2 + 1}{x^3} = x^3 + x + \frac{1}{x} + \frac{1}{x^3}$$

$$= \left(x^3 + \frac{1}{x^3}\right) + \left(x + \frac{1}{x}\right) = 198 + 6 = 204$$

7. If $x = 23$ then find the value of $x^4 - 24x^3 + 24x^2 - 24x$?

Sol. $x^4 - 24x^3 + 24x^2 - 24x$

$$= x^4 - 23x^3 - x^3 + 23x^2 + x^2 - 23x - x$$

$$= x^4 - x^4 - x^3 + x^3 + x^2 - x^2 - x = -23$$

8. If $x + \frac{1}{x} = 1$, then find the value

$$\text{of } x^{72} + x^{71} + x^{70} + x^{69} + x^{68} + x^{67} + x^{54}?$$

Sol. $x^{72} + x^{69} + x^{71} + x^{68} + x^{70} + x^{67} + x^{54}$

$$= x^{69}(x^3 + 1) + x^{68}(x^3 + 1) + x^{67}(x^3 + 1) + (x^3)^{18}$$

$$\text{Here, } x^3 + 1 = 0, \quad x^3 = -1$$

$$= x^{69} \times 0 + x^{68} \times 0 + x^{67} \times 0 + (-1)^{18} = 1$$

9. $x = 2 - 2^{1/3} + 2^{2/3}$, find $x^3 - 6x^2 + 18x + 28$?

Sol. $x = 2 + 2^{2/3} - 2^{1/3}$, $x - 2 = 2^{2/3} - 2^{1/3}$

Cubing both sides

$$(x - 2)^3 = (2^{2/3} - 2^{1/3})^3$$

$$x^3 - 8 - 6x^2 + 12x = 4 - 2 - 3 \times 2^{2/3} \times 2^{1/3} (2^{2/3} - 2^{1/3})$$

$$x^3 - 8 - 6x^2 + 12x = 2 - 3 \times 2 (x - 2)$$

$$x^3 - 6x^2 + 18x = 22$$

$$x^3 - 6x^2 + 18x + 28 = 22 + 28$$

$$x^3 - 6x^2 + 18x + 28 = 50$$

10. If $x^2 + 2 = 2x$, find $x^4 - x^3 + x^2 + 2$?

Sol. $x^2 + 2 = 2x$

Squaring both sides

$$(x^2 + 2)^2 = 4x^2$$

$$x^4 + 4 + 4x^2 = 4x^2, x^4 = -4$$

$$x^4 - x^3 - x^2 + 2 = -4 - x^3 + x^2 + 2$$

$$= -x^3 + x^2 - 2 = -x^2(x - 1) - 2$$

$$= -x^2 \left(\frac{x^2}{2} \right) - 2 = \frac{-x^4}{2} - 2 = \frac{4}{2} - 2 = 2 - 2 = 0$$

11. If $x + \frac{1}{x} = 9$ then find the value of $\frac{3x}{x^2 + 5x + 1}$?

Sol. $x + \frac{1}{x} = 9$, $\frac{x^2 + 1}{x} = 9$

$$x^2 + 1 = 9x$$

...(i)

$$\frac{3x}{x^2 + 5x + 1} = \frac{3x}{(x^2 + 1) + 5x}$$

From equation (i)

$$= \frac{3x}{9x + 5x} = \frac{3x}{14x} = \frac{3}{14}$$

12. If $\frac{x^2}{by + cz} = \frac{y^2}{cz + ax} = \frac{z^2}{ax + by} = 1$, then find the value

of $\frac{a}{a+x} + \frac{b}{b+y} + \frac{c}{c+z}$?

Sol. $\frac{x^2}{by + cz} = \frac{y^2}{cz + ax} = \frac{z^2}{ax + by} = 1$

therefore, $x^2 = by + cz$, $y^2 = cz + ax$, $z^2 = ax + by$

$$= \frac{a}{a+x} + \frac{b}{b+y} + \frac{c}{c+z} = \frac{ax}{ax+x^2} + \frac{by}{by+y^2} + \frac{cz}{cz+z^2}$$

$$= \frac{ax}{ax+by+cz} + \frac{by}{ax+by+cz} + \frac{cz}{ax+by+cz}$$

$$= \frac{ax+by+cz}{ax+by+cz} = 1$$

13. If $ax + by = 3$, $bx - ay = 4$ and $x^2 + y^2 = 1$, then find the value of $a^2 + b^2$?

Sol. If $ax + by = m$ and $bx - ay = n$ then

$$(a^2 + b^2)(x^2 + y^2) = m^2 + n^2$$

$$ax + by = 3 \quad \dots(i)$$

$$bx - ay = 4 \quad \dots(ii)$$

On squaring and adding both the equation

$$a^2x^2 + b^2y^2 + b^2x^2 + a^2y^2 = 9 + 16$$

$$a^2x^2 + a^2y^2 + b^2x^2 + b^2y^2 = 25$$

$$a^2(x^2 + y^2) + b^2(x^2 + y^2) = 25$$

$$(a^2 + b^2)(x^2 + y^2) = 25$$

$$a^2 + b^2 = 25$$

14. If $x + y + z = 10$, $x^2 + y^2 + z^2 = 30$, then find the value of $x^3 + y^3 + z^3 - 3xyz$?

Sol. $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$10^2 = 30 + 2(xy + yz + zx)$$

$$2(xy + yz + zx) = 100 - 30$$

$$2(xy + yz + zx) = 70$$

$$xy + yz + zx = 35$$

$$x^3 + y^3 + z^3 - 3xyz = 10 [30 - 35]$$

$$= 10(-5) = -50$$

15. If $x - y = \frac{x+y}{7} = \frac{xy}{4}$ then find the value of xy ?

Sol. $\frac{x-y}{1} = \frac{x+y}{7} = \frac{xy}{4}$

By using ratio proportion

$$\frac{x-y}{1} = \frac{x+y}{7} = \frac{xy}{4}$$

$$= \frac{(x-y) + (x+y)}{1+7} = \frac{(x-y) - (x+y)}{1-7}$$

$$= \frac{x-y}{1} = \frac{x+y}{4} = \frac{xy}{4} = \frac{x}{4} = \frac{y}{3}$$

Again, using ratio proportion

$$\frac{xy}{4} = \frac{x}{4} = \frac{y}{3} = \frac{\sqrt{xy}}{\sqrt{4 \times 3}}$$

or $\frac{(xy)^2}{16} = \frac{xy}{12}$, $xy = \frac{16}{12} = \frac{4}{3}$

16. If $\frac{x^3 + 3x}{3x^2 + 1} = \frac{189}{61}$ then find the value of x ?

Sol. $\frac{x^3 + 3x}{3x^2 + 1} = \frac{189}{61}$

By using componendo and dividendo

$$\frac{x^3 + 3x + 3x^2 + 1}{x^3 + 3x - (3x^2 + 1)} = \frac{189 + 61}{189 - 61} = \frac{125}{64}$$

$$\left(\frac{x+1}{x-1}\right)^3 = \left(\frac{5}{4}\right)^3, \quad \frac{x+1}{x-1} = \frac{5}{4}$$

$$4x + 4 = 5x - 5, \quad x = 9$$

17. If $a = \sqrt{7 + 2 \times \sqrt{4} \times \sqrt{3}}$ and $b = \sqrt{7 - 2\sqrt{12}}$, then find the value of $a^3 + b^3$?

Sol. $a = \sqrt{7 + 2 \times \sqrt{4} \times \sqrt{3}} = \sqrt{4 + 3 + 2 \times \sqrt{4} \times \sqrt{3}}$
 $= \sqrt{(2 + \sqrt{3})^2} = 2 + \sqrt{3}$

and, $b = \sqrt{7 - 2\sqrt{12}}$

Similarly, $b = 2 - \sqrt{3}$

$$a + b = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$$

$$ab = (2 + \sqrt{3})(2 - \sqrt{3})$$

$$ab = 2^2 - (\sqrt{3})^2 = 1$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$64 = a^3 + b^3 + 3(4)$$

$$a^3 + b^3 = 64 - 12 = 52$$

18. If $2^x = 3^y = 6^{-z}$ then find $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$?

Sol. Let $2^x = 3^y = 6^{-z} = K$

$$2^x = K, 2 = K^{1/x}, \quad 3^y = K, 3 = K^{1/y}$$

$$6^{-z} = K, 6 = K^{-1/z}, \quad 2 \times 3 = 6$$

$$K^{1/x} \cdot K^{1/y} = K^{-1/z}, \quad K^{(1/x + 1/y)} = K^{-1/z}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{-1}{z}, \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

19. Given that $a + b + c = 2$ and $ab + bc + ca = 1$ then find the value of $(a + b)^2 + (b + c)^2 + (c + a)^2$.

Sol. Given, $a + b + c = 2$

and, $ab + bc + ca = 1$

Now,

$$\begin{aligned} (a + b)^2 + (b + c)^2 + (c + a)^2 &= a^2 + b^2 + 2ab + b^2 + c^2 + 2bc + c^2 + a^2 + 2ac \\ &= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + a^2 + b^2 + c^2 \\ &= (a + b + c)^2 + a^2 + b^2 + c^2 \\ &= (a + b + c)^2 + (a + b + c)^2 - 2(ab + bc + ca) \\ &= 2(a + b + c)^2 - 2(ab + bc + ca) \\ &= 2 \times 2^2 - 2 \times 1 \\ &= 8 - 2 = 6 \end{aligned}$$

20. If $\frac{x}{xa + yb + zc} = \frac{y}{ya + zb + xc} = \frac{z}{za + xb + yc}$

and $x + y + z \neq 0$ then find each ratio?

Sol. $\frac{x}{xa + yb + zc} = \frac{y}{ya + zb + xc} = \frac{z}{za + xb + yc} = K$

$$x = K(xa + yb + zc)$$

$$y = K(ya + zb + xc)$$

and $z = K(za + xb + yc)$

Now,

$$x + y + z = K[a(x + y + z) + b(x + y + z) + c(x + y + z)]$$

$$x + y + z = K[(a + b + c)(x + y + z)]$$

$$1 = K(a + b + c)$$

$$K = \frac{1}{a + b + c}$$

Foundation

Questions

1. If $x * y = (x + 3)^2(y - 1)$, then the value of $5 * 4$ is

- (a) 192 (b) 182
(c) 180 (d) 172

2. $a \times b = a^b$, then the value of 5×3 is

- (a) 125 (b) 243
(c) 53 (d) 15

3. If $1.5a = 0.04b$ then $\frac{b-a}{b+a}$ is equal to

- (a) $\frac{73}{77}$ (b) $\frac{77}{73}$

(c) $\frac{2}{75}$

(d) $\frac{75}{2}$

4. If $a = 11$ and $b = 9$, then the value of $\left(\frac{a^2 + b^2 + ab}{a^3 - b^3}\right)$ is

(a) $\frac{1}{2}$

(b) 2

(c) $\frac{1}{20}$

(d) 20

5. If $a^4 + b^4 = a^2b^2$, then $(a^6 + b^6)$ equal to

(a) 0

(b) 1

(c) $a^2 + b^2$

(d) $a^2b^4 + a^4b^2$

6. If $x : y = 2 : 1$, then $(5x^2 - 13xy + 6y^2)$ is equal to-

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) 0 (d) $\frac{55}{4}$

7. The expression $x^4 - 2x^2 + K$ will be a perfect square when the value of K is

- (a) 2 (b) 1
(c) -1 (d) -2

8. If $a = 1.21$, $b = 2.12$ and $c = -3.33$ then, the value of $a^3 + b^3 + c^3 - 3abc$ is

- (a) 0 (b) 1
(c) 2 (d) 3

9. If $x * y = x^2 + y^2 - xy$, then the value of $9 * 11$ is

- (a) 93 (b) 103
(c) 113 (d) 121

10. If $p = 999$, then the value of $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ is

- (a) 1000 (b) 999
(c) 998 (d) 1002

11. If $X * Y = X^2 + Y^2 - XY$ then $11 * 13$ is

- (a) 117 (b) 147
(c) 119 (d) 433

12. If $a^3b = abc = 180$, a, b, c are positive integers, then the value of c is

- (a) 110 (b) 1
(c) 4 (d) 25

13. If $x = -1$, then the value of

- $\frac{1}{x^{99}} + \frac{1}{x^{98}} + \frac{1}{x^{97}} + \frac{1}{x^{96}} + \frac{1}{x^{95}} + \frac{1}{x^{94}} + \frac{1}{x} - 1$ is
(a) 1 (b) 0
(c) -2 (d) -1

14. If $\frac{4x}{3} + 2P = 12$ for what value of P , $x = 6$?

- (a) 6 (b) 4
(c) 2 (d) 1

15. If $a^{\frac{1}{3}} = 11$ then the value of $a^2 - 331a$ is

- (a) 1331331 (b) 1331000
(c) 1334331 (d) 1330030

16. If $x + \frac{1}{x} = 2$, $x \neq 0$, then value of $x^2 + \frac{1}{x^3}$ is equal to

- (a) 1 (b) 2
(c) 3 (d) 4

17. If $x^2 - 3x + 1 = 0$, then the value of $x^3 + \frac{1}{x^3}$ is

- (a) 9 (b) 18
(c) 27 (d) 1

18. If $m + \frac{1}{m-2} = 4$, find the value of $(m-2)^2 + \frac{1}{(m-2)^2}$.

- (a) -2 (b) 0
(c) 2 (d) 4

19. If $\left(a + \frac{1}{a}\right)^2 = 3$, then $a^3 + \frac{1}{a^3}$ is

- (a) -2 (b) 0
(c) 2 (d) 4

20. If $\frac{a}{b} + \frac{b}{a} - 1 = 0$, then the value of $a^3 + b^3$ is

- (a) 3 (b) 0
(c) 1 (d) -1

21. If $(x-3)^2 + (y-5)^2 + (z-4)^2 = 0$, then the value of

- $\frac{x^2}{9} + \frac{y^2}{25} + \frac{z^2}{16}$ is
(a) 12 (b) 9
(c) 3 (d) 1

22. If $\frac{a}{b} = \frac{2}{3}$ and $\frac{b}{c} = \frac{4}{5}$, then the ratio $\frac{a+b}{b+c}$ equal to:

- (a) $\frac{20}{27}$ (b) $\frac{27}{20}$
(c) $\frac{6}{8}$ (d) $\frac{8}{6}$

23. If $y : x = 4 : 15$, then the value of $\frac{(x-y)}{(x+y)}$ is

- (a) $\frac{11}{19}$ (b) $\frac{19}{11}$
(c) $\frac{4}{11}$ (d) $\frac{15}{19}$

24. If $\frac{3x+2y}{3x-2y} = \frac{4}{3}$ then find the value of $\frac{x^2+y^2}{x^2-y^2}$

- (a) $\frac{185}{205}$ (b) $\frac{205}{187}$
(c) $\frac{14}{3}$ (d) None of these

25. If $x + \frac{1}{x} = 5$ then $\frac{3x}{2x^2+2-5x}$ will be:

- (a) $\frac{2}{3}$ (b) 0
(c) 1 (d) $\frac{3}{5}$

26. If $\left(a + \frac{1}{a}\right)^2 = 3$ then $a^3 + \frac{1}{a^3} + 3\sqrt{3}$ will be

(a) $\frac{10\sqrt{3}}{3}$ (b) 0

(c) $3\sqrt{3}$ (d) $6\sqrt{3}$

27. If $P = \frac{x^2 - 36}{x^2 - 49}$ and $Q = \frac{x+6}{x+7}$ then the value of $\frac{P}{Q}$ is:

(a) $\frac{x-6}{x-7}$ (b) $\frac{x-6}{x+7}$

(c) $\frac{x-7}{x+6}$ (d) $\frac{x+6}{x-7}$

28. If $x + \frac{1}{x} = 2$, then find the value of $x^{17} + \frac{1}{x^{19}}$?

(a) 0 (b) 1

(c) -1 (d) 2

29. Simplify $\frac{2x^4 - 162}{(x^2 + 9)(2x - 6)}$

(a) $\frac{x^2 - 9}{x^2 + 9}$ (b) $x + 3$

(c) $x - 3$ (d) $\frac{x+3}{x-3}$

30. $\frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2bc - c^2}$ is equal to:

(a) $\frac{a+b+c}{a-b+c}$ (b) $\frac{a-b-c}{a+b-c}$

(c) $\frac{a-b-c}{a-b+c}$ (d) $\frac{a-b+c}{a+b+c}$

31. If the average of x and $\frac{1}{x}$ be 1, then the value of

$8x^{10} + \frac{4}{x^5}$ will be.

(a) 12 (b) -12

(c) 0 (d) 1

32. If $x + y + z = 0$, then $(x+y)(y+z)(z+x)$ is equal to which of the following?

(a) $-xyz$ (b) $x^2 + y^2 + z^3$

(c) $x^3 + y^3 + z^3 + 3xyz$ (d) xyz

33. If $x^2 - 4x + 1 = 0$, then what is the value of $x^3 + \frac{1}{x^3}$?

(a) 44 (b) 48

(c) 52 (d) 64

34. If $x + y + z = 6$ and $xy + yz + zx = 11$ then what is the value $x^3 + y^3 + z^3 - 3xyz$?

(a) 18 (b) 36

(c) 54 (d) 66

35. If $a = 258$, $b = 260$ and $c = 262$ then value of $a^3 + b^3 + c^3 - 3abc$ is

(a) 9360 (b) 6240

(c) 7040 (d) 10560

36. If $pq + qr + rp = 0$ then what is the value of

$\frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - rp} + \frac{r^2}{r^2 - pq}$?

(a) 0 (b) 1

(c) -1 (d) 3

37. If u, v, w are real numbers such that

$u^3 - 8v^3 - 27w^3 = 18uvw$, which one of the following is true?

(a) $u - v + w = 0$

(b) $u = -v = -w$

(c) $u - 2v = 3w$

(d) $u + 2v = -3w$

38. If $(x^2 + y^2)(p^2 + q^2) = (xp + yq)^2$ then

(a) $xy = pq$

(b) $px = yp$

(c) $xq = yp$

(d) None of these

39. If $a + b + c = p$, $abc = q$ and $ab + bc + ca = 0$ then what is the value of $a^2b^2 + b^2c^2 + c^2a^2$?

(a) $2pq$

(b) $-2pq$

(c) $3pq$

(d) $-3pq$

40. If $x = \frac{p+q}{p-q}$ and $y = \frac{p-q}{p+q}$ then the value of $\frac{x-y}{x+y}$ is

(a) $\frac{p^2 + q^2}{2pq}$

(b) $\frac{2pq}{p^2 + q^2}$

(c) $\frac{2pq}{p^2 - q^2}$

(d) $\frac{2(p^2 - q^2)}{pq}$

Moderate

1. If the sum of $\frac{a}{b}$ and its reciprocal is 1 and $a \neq 0, b \neq 0$ then the value of $a^3 + b^3 + 2$ is
 (a) 2 (b) -1
 (c) 0 (d) 1
2. If $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 4$, then the value of $x^2 + y^2$ is
 (a) 2 (b) 4
 (c) 8 (d) 16
3. If $x^2 = y + z, y^2 = z + x, z^2 = x + y$, then the value of $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$ is
 (a) -1 (b) 1
 (c) 2 (d) 4
4. If $x + \frac{1}{x} = \sqrt{3}$ then the value of $x^{18} + x^{12} + x^6 + 1$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
5. If $a^2 + b^2 = 2$ and $c^2 + d^2 = 1$, then the value of $(ad - bc)^2 + (ac + bd)^2$ is
 (a) $\frac{4}{9}$ (b) $\frac{1}{2}$
 (c) 1 (d) 2
6. If $x + \frac{2}{3 + \frac{4}{5 + \frac{7}{6}}} = 10$, then the value of x
 (a) $\frac{1276}{135}$ (b) $\frac{53}{6}$
 (c) 4.35 (d) 9
7. If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = 3$, then $\frac{2a^2 + 3c^2 + 4e^2}{2b^2 + 3d^2 + 4f^2} = ?$
 (a) 2 (b) 3
 (c) 4 (d) 9
8. If $a + \frac{1}{a} = 6$ then find the value of $a^4 + \frac{1}{a^4}$.
 (a) 1154 (b) 1158
 (c) 1164 (d) 1168
9. If $x^4 + \frac{1}{x^4} = 23$, then find the value of $\left(x - \frac{1}{x}\right)^2$
 (a) 7 (b) -7
 (c) -3 (d) 3
10. If $x^3 + y^3 = 35$ and $x + y = 5$, find the value of $\left(\frac{1}{x} + \frac{1}{y}\right)^3$
 (a) $\frac{3}{7}$ (b) $\frac{3}{8}$
 (c) $\frac{5}{6}$ (d) $\frac{13}{7}$
11. If $x^2 + y^2 + z^2 = 2(x - y - z) - 3$, then the value of $5x - 4y + 2z$
 (a) 6 (b) 8
 (c) 7 (d) 9
12. If $2p + \frac{1}{p} = 4$, the value of $p^3 + \frac{1}{8p^3}$ is
 (a) 4 (b) 5
 (c) 8 (d) 15
13. If $x = 3 + \sqrt{8}$, the value of $\left(x^2 + \frac{1}{x^2}\right)$ is
 (a) $32\sqrt{2}$ (b) 34
 (c) 38 (d) $40\sqrt{2}$
14. If $\frac{2a+b}{a+4b} = 3$, then find the value of $\frac{a+b}{a+2b}$
 (a) $\frac{5}{9}$ (b) $\frac{2}{7}$
 (c) $\frac{10}{9}$ (d) $\frac{10}{7}$
15. If $a = 4.965, b = 2.343$ and $c = 2.622$, then the value of $a^3 - b^3 - c^3 - 3abc$ is :
 (a) -2 (b) 8
 (c) 0 (d) 9.9
16. If $*$ is an operation such that $x * y = 3x + 2y$, then $2 * 3 + 3 * 4$ is equal to:
 (a) 18 (b) 29
 (c) 32 (d) 38
17. The value of the expression $x^4 - 17x^3 + 17x^2 - 17x + 17$ at $x = 16$ is
 (a) 0 (b) 1
 (c) 2 (d) 3

18. If $a + b + c + d = 1$, then the maximum value of $(1 + a)(1 + b)(1 + c)(1 + d)$ is

- (a) 1
(b) $\left(\frac{1}{2}\right)^3$
(c) $\left(\frac{3}{4}\right)^3$
(d) $\left(\frac{5}{4}\right)^4$

19. If $\frac{1}{\sqrt[3]{4} + \sqrt[3]{2} + 1} = a\sqrt[3]{4} + b\sqrt[3]{2} + c$ and a, b, c , are rational numbers, then $a + b + c$ is equal to

- (a) 0
(b) 1
(c) 2
(d) 3

20. Out of the given responses, one of the factors of $(a^2 - b^2) + (b^2 - c^2) + (c^2 - a^2)$

- (a) $(a + b)(a - b)$
(b) $(a + b)(a + b)$
(c) $(a - b)(a - b)$
(d) $(a - c)(a - c)$

21. If $p = 99$, then value of $p(p^2 + 3p + 3)$ is

- (a) 999
(b) 9999
(c) 99999
(d) 999999

22. If $P = 123$, then the value $\sqrt[3]{p(p^2 + 3p + 3) + 1} = ?$

- (a) 5
(b) 7
(c) 123
(d) 124

23. The expression $x^4 - 4x^2 + k$ will be a perfect square when the value of k is

- (a) 2
(b) 1
(c) -1
(d) 4

24. If $x = 7 - 4\sqrt{3}$ then the value of $\left(x + \frac{1}{x}\right)$ is:

- (a) $3\sqrt{3}$
(b) $8\sqrt{3}$
(c) $14 + 8\sqrt{3}$
(d) 14

25. If $x = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ and $y = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$ the value of $x^2 + y^2$ is:

- (a) 14
(b) 13
(c) 15
(d) 10

26. If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$ then the value of

$$\frac{a^2 + ab + b^2}{a^2 - ab + b^2}$$

- (a) $\frac{3}{4}$
(b) $\frac{4}{3}$
(c) $\frac{3}{5}$
(d) $\frac{5}{3}$

27. If $x - \frac{1}{x} = 4$, then $\left(x + \frac{1}{x}\right)$ is equal to

- (a) $2\sqrt{2}$
(b) $2\sqrt{5}$
(c) $4\sqrt{2}$
(d) $4\sqrt{5}$

28. If $x = 3 + 2\sqrt{2}$, then the value of $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)$ is:

- (a) 1
(b) 2
(c) $2\sqrt{2}$
(d) $3\sqrt{2}$

29. If $x + \frac{1}{2x} = 2$, find the value of $8x^3 + \frac{1}{x^3}$.

- (a) 48
(b) 88
(c) 40
(d) 44

30. If $\left(x + \frac{1}{x}\right)^2 = 3$, then the value of $(x^{72} + x^{66} + x^{54} + x^{48} + x^{36} + x^{30} + x^{24} + x^{18} + x^6 + 1)$ is

- (a) 0
(b) 2
(c) 3
(d) 4

31. If $n = 7 + 4\sqrt{3}$, then the value of $\left(\sqrt{n} + \frac{1}{\sqrt{n}}\right)$ is

- (a) $2\sqrt{3}$
(b) 4
(c) $2\sqrt{3}$
(d) $-2\sqrt{3}$

32. If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b}$, then which one is true?

- (a) $\frac{x-y}{b-a} = \frac{y-z}{c-b} = \frac{z-x}{a-c}$
(b) $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$
(c) $\frac{x-y}{c} = \frac{y-z}{b} = \frac{z-x}{c}$
(d) None of these

33. If $x^2 + y^2 + z^2 = 2(x - y - z) - 3$, then the value of $2x - 3y + 4z$ is: [x, y, z are all real numbers]

- (a) 9
(b) 1
(c) 3
(d) 0

34. If $x + y + z = 6$ and $x^2 + y^2 + z^2 = 20$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

- (a) 64
(b) 70
(c) 72
(d) 76

35. If $x + \frac{1}{x} = 99$, find the value of $\frac{100x}{2x^2 + 102x + 2}$

- (a) $\frac{1}{6}$
(b) $\frac{1}{2}$
(c) $\frac{1}{3}$
(d) $\frac{1}{4}$

36. If $x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$, then the value of $x^2 + \frac{1}{x^2}$ is

- (a) $2\frac{1}{9}$ (b) $2\frac{4}{9}$
(c) $3\frac{1}{9}$ (d) $3\frac{4}{9}$

37. If $x^2 - 3x + 1 = 0$, then the value of $x^2 + x + \frac{1}{x} + \frac{1}{x^2}$ is

- (a) 10 (b) 2
(c) 6 (d) 8

38. If $(x - a)(x - b) = 1$ and $a - b + 5 = 0$, then the value of

$(x - a)^3 - \frac{1}{(x - a)^3}$ is:

- (a) -125 (b) 1
(c) 125 (d) 140

39. If $x : y = 3 : 2$, then the ratio $2x^2 + 3y^2 : 3x^2 - 2y^2$ is equal to:

- (a) 12 : 5 (b) 6 : 5
(c) 30 : 19 (d) 5 : 3

40. If $A : B = 1 : 2$, $B : C = 3 : 4$ and $C : D = 5 : 6$, find $D : C : B : A$?

- (a) 6 : 5 : 4 : 2 (b) 6 : 3 : 2 : 1
(c) 6 : 4 : 2 : 1 (d) 48 : 40 : 30 : 15

Difficult

1. If $\frac{\sqrt{x+4} + \sqrt{x-4}}{\sqrt{x+4} - \sqrt{x-4}} = 2$, then x is equal to:

- (a) 2.4 (b) 3.2
(c) 4 (d) 5

2. If $x = \sqrt[3]{a + \sqrt{a^2 + b^3}} + \sqrt[3]{a - \sqrt{a^2 + b^3}}$, then $x^3 + 3bx$ is equal to:

- (a) 0 (b) a
(c) $2a$ (d) 1

3. If $\frac{4\sqrt{3} + 5\sqrt{2}}{\sqrt{48} + \sqrt{18}} = a + b\sqrt{6}$ then the values of a and b are respectively:

- (a) $\frac{9}{15}, \frac{4}{15}$ (b) $\frac{3}{11}, \frac{4}{33}$
(c) $\frac{9}{10}, \frac{2}{5}$ (d) $\frac{3}{5}, \frac{4}{15}$

4. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$, then $c + \frac{1}{a}$ is equal to:

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

5. If $a + b + c = 0$, then the value of $\frac{a^2 + b^2 + c^2}{a^2 - bc}$ is

- (a) 0 (b) 1
(c) 2 (d) 3

6. If $x + \frac{1}{x} = 3$, then the value of $\left(x^5 + \frac{1}{x^5}\right)$ is

- (a) 322 (b) 126
(c) 123 (d) 113

7. If $\frac{xy}{x+y} = a$, $\frac{xz}{x+z} = b$ and $\frac{yz}{y+z} = c$ where a, b, c are all non-zero numbers, then x equals to:

- (a) $\frac{2abc}{ab + bc - ac}$ (b) $\frac{2abc}{ab + ac - bc}$
(c) $\frac{2abc}{ac + bc - ab}$ (d) $\frac{2abc}{ab + bc - ac}$

8. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, then $\frac{18c^2 - 7a^2}{45c^2 + 20a^2}$ is equal to:

- (a) $\frac{1}{3}$ (b) $\frac{2}{5}$
(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

9. $x^a \cdot x^b \cdot x^c = 1$ then $a^3 + b^3 + c^3$ is equal to:

- (a) 9 (b) abc
(c) $a + b + c$ (d) $3abc$

10. If $x = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ then the value of $5x^2 - 5x - 1$ will be:

- (a) 0 (b) 3
(c) 4 (d) 5

11. If α and β are the roots of the equation $x^2 + px + q = 0$

then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ will be:

- (a) $\frac{p^2 - 2q}{q}$ (b) $\frac{p^2 + 2q}{q}$
(c) $\frac{-p^2 - 2q}{q}$ (d) $\frac{-p^2 + 2q}{q}$

12. If $xy(x+y) = 1$ then $\frac{1}{x^3y^3} - x^3 - y^3$ is equal to:
 (a) 0 (b) 1
 (c) 3 (d) -2
13. If $ax + by = 6$, $bx - ay = 2$ and $x^2 + y^2 = 4$ then the value of $(a^2 + b^2)$ would be.
 (a) 2 (b) 4
 (c) 5 (d) 10
14. If $2a - \frac{2}{a} + 3 = 0$, the value of $\left(a^3 - \frac{1}{a^3} + 2\right)$ is
 (a) 5 (b) $-\frac{35}{8}$
 (c) $-\frac{40}{7}$ (d) $-\frac{47}{8}$
15. For what value of a is $2x^3 + ax^2 + 11x + a + 3$ exactly divisible by $(2x - 1)$?
 (a) -4 (b) -5
 (c) -6 (d) -7
16. If both $(x - 2)$ and $\left(x - \frac{1}{2}\right)$ are factors of $px^2 + 5x + r$ then which one is correct:
 (a) $p = 2r$ (b) $pr = 1$
 (c) $\frac{p}{r} = 1$ (d) None of these
17. If $a + b + c = 2s$, then find the value of $(s-a)^3 + (s-b)^3 + 3(s-a)(s-b)c$:
 (a) c (b) c^2
 (c) c^3 (d) $2c^2$
18. If $x(x+y+z) = 28$, $y(x+y+z) = 70$ and $z(x+y+z) = 98$ then the value of $3(x+y+z)$ is:
 (a) -28 (b) +24
 (c) 42 (d) 36
19. If $a^x = (x+y+z)^y$ and $a^y = (x+y+z)^z$ and $a^z = (x+y+z)^x$, then the value of $x+y+z$ (given $a \neq 0$) is:
 (a) 0 (b) a^3
 (c) 1 (d) a
20. If $\frac{5x-7y+10}{1} = \frac{3x+2y+1}{8} = \frac{11x+4y-10}{9}$, then $x+y$ equals
 (a) 1 (b) 2
 (c) 3 (d) -3
21. If average of x and $\frac{1}{x}$ ($x \neq 0$) is M then what is the average of x^2 and $\frac{1}{x^2}$?
 (a) $1 - M^2$ (b) $1 - 2M$
 (c) $2M^2 - 1$ (d) $2M^2 + 1$
22. If $\frac{x}{(b-c)(b+c-2a)} = \frac{y}{(c-a)(c+a-2b)} = \frac{z}{(a-b)(a+b-2c)}$ then $x+y+z$ is equal to:
 (a) $a+b+c$ (b) $a^2+b^2+c^2$
 (c) 0 (d) $(a+b+c)^2$
23. Which one of the following is not a factor of $x^8 + x^4 + 1$?
 (a) $x - \sqrt{x} + 1$ (b) $x^2 - x + 1$
 (c) $x^4 - x^2 + 1$ (d) $x^2 - 2x + 1$
24. If $pqr = 1$, then what is value of the expression $\frac{1}{1+p+q^{-1}} + \frac{1}{1+q+r^{-1}} + \frac{1}{1+r+p^{-1}}$?
 (a) 1 (b) -1
 (c) 0 (d) $\frac{1}{3}$
25. If $x+y+z=0$, then $\left[\frac{y-z-x}{2}\right]^3 + \left[\frac{z-x-y}{2}\right]^3 + \left[\frac{x-y-z}{2}\right]^3$ equal to:
 (a) $24xyz$ (b) $-24xyz$
 (c) $3xyz$ (d) xyz

Previous Year (Memory Based)

1. If $x^4 + \frac{1}{x^4} = 23$ then what is the value of $\left(x - \frac{1}{x}\right)^2$?
 (a) -3 (b) 3
 (c) 7 (d) -7
2. If $a+b=6$, $a-b=2$ then what is the value of $2(a^2+b^2)$?
 (a) 20 (b) 30
 (c) 40 (d) 10
3. If x is real and $x^4 + \frac{1}{x^4} = 119$, then value of $\left(x - \frac{1}{x}\right)$ is:
 (a) ± 4 (b) ± 9
 (c) ± 3 (d) ± 2
4. If $a+b+c=8$, then the value of $(a-4)^3 + (b-3)^3 + (c-1)^3 - 3(a-4)(b-3)(c-1)$ is:
 (a) 2 (b) 4
 (c) 1 (d) 0

5. If $x = \sqrt{a} + \frac{1}{\sqrt{a}}$, $y = \sqrt{a} - \frac{1}{\sqrt{a}}$, then value of $x^4 + y^4 - 2x^2y^2$ is:
 (a) 16 (b) 20
 (c) 10 (d) 5
6. If $5a + \frac{1}{3a} = 5$, then value of $9a^2 + \frac{1}{25a^2}$ is:
 (a) $\frac{51}{5}$ (b) $\frac{29}{5}$
 (c) $\frac{52}{5}$ (d) $\frac{39}{5}$
7. If $x + \frac{1}{x} = 5$ then what is the value of $\frac{x^4 + \frac{1}{x^2}}{x^2 - 3x + 1}$?
 (a) 70 (b) 50
 (c) 110 (d) 55
8. If $(5x^2 - 3y^2) : xy = 11 : 2$ then what is the positive value of $\frac{x}{y}$?
 (a) $\frac{5}{2}$ (b) $\frac{3}{2}$
 (c) $\frac{5}{3}$ (d) $\frac{7}{2}$
9. If $a^3 - b^3 = 56$ and $a - b = 2$ then what is the value of $(a^2 + b^2)$?
 (a) -12 (b) 20
 (c) 18 (d) -10
10. If $x = 2t$ and $y = \frac{2t-1}{3}$, then for what value of t , $x = y$ is correct?
 (a) $\frac{1}{3}$ (b) $-\frac{1}{4}$
 (c) $\frac{1}{4}$ (d) $-\frac{1}{2}$
11. If x and y are two real numbers and $x + y = 8$, then maximum value of xy is:
 (a) 16 (b) 18
 (c) 12 (d) 15
12. If $(2 + \sqrt{3})a = (2 - \sqrt{3})b = 1$, then the value of $\frac{1}{a} + \frac{1}{b}$ is
 (a) 1 (b) 2
 (c) $2\sqrt{3}$ (d) 4
13. If $3x + \frac{3}{x} = 1$, then $x^3 + \frac{1}{x^3} + 1$ is
 (a) 52 (b) 64
 (c) 4 (d) 25
14. If $4x + 5y = 83$ and $3x : 2y = 21 : 22$, then $y - x$ equals
 (a) 3 (b) 4
 (c) 7 (d) 11
15. The factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ are:
 (a) $(a - 2b - 4)(a - 2b + 2)$
 (b) $(a - b + 2)(a - 4b - 4)$
 (c) $(a + 2b - 4)(a + 2b + 2)$
 (d) $(a + 2b - 1)(a - 2b + 1)$
16. If $p = \frac{5}{18}$, then the value of $27p^3 - \frac{1}{216} - \frac{9}{2}p^2 + \frac{1}{4}p$ is:
 (a) $\frac{4}{27}$ (b) $\frac{5}{27}$
 (c) $\frac{8}{27}$ (d) $\frac{10}{27}$
17. If $x + \frac{1}{x} = 1$, ($x \neq 0$), then the value of $\frac{x^2 + 3x + 1}{x^2 + 7x + 1}$ is:
 (a) 1 (b) 2
 (c) 0 (d) $\frac{1}{2}$
18. If $a = \sqrt{6} + \sqrt{5}$ and $b = \sqrt{6} - \sqrt{5}$, then $2a^2 - 5ab + 2b^2$ is equal to:
 (a) 38 (b) 39
 (c) 40 (d) 41
19. Simplify $\sqrt{1 + \left(\frac{x^4 - 1}{-2x^2}\right)^2}$
 (a) $\frac{x^4 + 1}{2x^2}$ (b) $\frac{\sqrt{x^2 + 1}}{2x^2}$
 (c) $\frac{x^4 + 2x^2 - 1}{2x^2}$ (d) $\frac{x^4 - 1}{2x^4}$
20. If $x^2 + 4x + 3 = 0$, then the value of $\frac{x^3}{x^6 + 27x^3 + 27}$ is:
 (a) -1 (b) $-\frac{1}{2}$
 (c) 1 (d) $\frac{1}{2}$
21. If $x + y = 4$ and $\frac{1}{x} + \frac{1}{y} = 4$, then the value of $x^3 + y^3$ is:
 (a) 52 (b) 64
 (c) 4 (d) 25

22. If $\frac{x}{xa+yb+zc} = \frac{y}{ya+zb+xc} = \frac{z}{za+xb+yc}$ and $x+y+z \neq 0$, then each ratio is:
- (a) $\frac{1}{a-b-c}$ (b) $\frac{1}{a+b-c}$
(c) $\frac{1}{a-b+c}$ (d) $\frac{1}{a+b+c}$
23. If $x = 3 + 2\sqrt{2}$, then the values of $x^3 + \frac{1}{x^3}$ and $x^3 - \frac{1}{x^3}$ are respectively:
- (a) $140\sqrt{2}, 198$ (b) $234, 216$
(c) $216, 234$ (d) $198, 140\sqrt{2}$
24. If $x^3 + y^3 = 9$ and $x + y = 3$, then the value of $x^4 + y^4$ is:
- (a) 81 (b) 32
(c) 27 (d) 17
25. If $x = p + \frac{1}{p}$ and $y = p - \frac{1}{p}$, then value of $x^4 - 2x^2y^2 + y^4$ is:
- (a) 24 (b) 4
(c) 16 (d) 8
26. If $x + \frac{1}{2x} = 2$, then find the value of $8x^3 + \frac{1}{x^3}$.
- (a) 48 (b) 88
(c) 40 (d) 44
27. If $\frac{x}{2x^2+5x+2} = \frac{1}{6}$, then the value of $\left(x + \frac{1}{x}\right)$ is:
- (a) 2 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) -2
28. If $a + b + c = 0$, then the value of $\frac{a^2 + b^2 + c^2}{a^2 - bc} - 2$ is
- (a) 0 (b) 1
(c) 2 (d) 3
29. If $5^{\sqrt{x}} + 12^{\sqrt{x}} = 13^{\sqrt{x}}$, then x is equal to:
- (a) $\frac{25}{4}$ (b) 4
(c) 9 (d) 6
30. If $x^2 + 2 = 2x$, then the value of $x^4 - x^3 + x^2 + 2$ is:
- (a) 0 (b) 1
(c) -1 (d) $\sqrt{2}$
31. If $a : b = 2 : 3$ and $b : c = 4 : 5$, find $a^2 : b^2 : bc$?
- (a) $4 : 9 : 45$ (b) $16 : 36 : 45$
(c) $16 : 36 : 25$ (d) $4 : 36 : 37$
32. The value of $\frac{(243)^{\frac{n}{5}} \cdot 3^{2n+1}}{9^n \cdot 3^{n-1}}$ is:
- (a) 1 (b) 9
(c) 3 (d) 3^n
33. If $x^2 - 3x + 1 = 0$, then the value of $x^3 + \frac{1}{x^3} - 9$ is:
- (a) 9 (b) 18
(c) 27 (d) 1
34. If $a^2 + b^2 + 2b + 4a + 5 = 0$, then the value of $\frac{a-b}{a+b}$ is:
- (a) 3 (b) -3
(c) $\frac{1}{3}$ (d) $-\frac{1}{3}$
35. If $x = 5 + 2\sqrt{6}$, then the value of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ is:
- (a) $2\sqrt{2}$ (b) $3\sqrt{2}$
(c) $2\sqrt{3}$ (d) $3\sqrt{3}$
36. $\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - 1\right)\left(x^2 + \frac{1}{x^2} + 1\right)$ is equal to:
- (a) $x^6 + \frac{1}{x^6}$ (b) $x^8 + \frac{1}{x^8}$
(c) $x^8 - \frac{1}{x^8}$ (d) $x^6 - \frac{1}{x^6}$
37. If $2p + \frac{1}{p} = 4$, then the value of $p^2 + \frac{1}{4p^2}$ is:
- (a) 4 (b) 3
(c) 8 (d) 15
38. If $\frac{2a+b}{a+4b} = 3$, then find the value of $\frac{a-b}{a+2b}$.
- (a) $\frac{5}{9}$ (b) $\frac{2}{7}$
(c) $\frac{4}{3}$ (d) $\frac{10}{7}$
39. If $x^{\frac{1}{3}} + y^{\frac{1}{3}} - z^{\frac{1}{3}} = 0$, then $\{(x+y-z)^3 + 27xyz\}$ is equal to:
- (a) -1 (b) 1
(c) 0 (d) 27
40. Given that, $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is
- (a) 1.45 (b) 1.88
(c) 2.9 (d) 3.7

Foundation

Solutions

1. (a); $5 * 4 = (5 + 3)^2 (4 - 1) = (8^2) (3)$
 $= 64 \times 3 = 192$

2. (a); $a \times b = a^b$
 $\therefore 5 \times 3 = 5^3 = 125$

3. (a); $1.5a = 0.04b$

$$\frac{b}{a} = \frac{1.5}{0.04} = \frac{150}{4}$$

By componendo and dividendo

$$\frac{b-a}{b+a} = \frac{150-4}{150+4} = \frac{146}{154} = \frac{73}{77}$$

4. (a); $\frac{a^2 + b^2 + ab}{a^3 - b^3} = \frac{a^2 + b^2 + ab}{(a-b)(a^2 + b^2 + ab)}$

$$= \frac{1}{a-b} = \frac{1}{11-9} = \frac{1}{2}$$

5. (a); $a^4 + b^4 - a^2b^2 = 0$... (i)

Now,

$$a^6 + b^6 = (a^2)^3 + (b^2)^3$$

$$= (a^2 + b^2)(a^4 + b^4 - a^2b^2)$$

From equation (i)

$$a^6 + b^6 = 0$$

6. (c); $\frac{x}{y} = 2 \Rightarrow x = 2y$

$$x - 2y = 0 \quad \dots (i)$$

$$5x^2 - 13xy + 6y^2$$

$$= 5x^2 - 10xy - 3xy + 6y^2$$

$$= 5x(x - 2y) - 3y(x - 2y)$$

$$= (x - 2y)(5x - 3y) = 0 \times (5x - 3y) = 0$$

7. (b); $(a - b)^2 = a^2 - 2ab + b^2$

$$x^4 - 2x^2 + K = (x^2)^2 - 2 \times x^2 \times 1 + K$$

$$\therefore K = (1)^2 = 1$$

8. (a); Here, $a = 1.21$, $b = 2.12$ and $c = -3.33$

$$a + b + c = 1.21 + 2.12 - 3.33$$

$$= 3.33 - 3.33 = 0$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{Hence, } a^3 + b^3 + c^3 - 3abc = 0$$

9. (b); $x * y = x^2 + y^2 - xy$

$$9 * 11 = 9^2 + 11^2 - 9 \times 11$$

$$= 81 + 121 - 99 = 202 - 99 = 103$$

10. (a); $p = 999$ (Given)

$$\text{Now, } \sqrt[3]{p(p^2 + 3p + 3)} + 1$$

$$= \sqrt[3]{p^3 + 3p^2 + 3p + 1} = \sqrt[3]{(p+1)^3}$$

$$= p + 1 = 999 + 1 = 1000$$

11. (b); $X * Y = X^2 + Y^2 - XY$

$$11 * 13 = 11^2 + 13^2 - 11 \times 13$$

$$= 121 + 169 - 143 = 290 - 143 = 147$$

12. (b); $a^3b = abc \Rightarrow a^2 = c$

$$\therefore a^3b = abc = 180$$

$$= 1^2 \times 180 \times 1 = 1^3 \times 180$$

$$\Rightarrow c = 1$$

13. (c); $\frac{1}{x^{99}} + \frac{1}{x^{98}} + \frac{1}{x^{97}} + \frac{1}{x^{96}} + \frac{1}{x^{95}} + \frac{1}{x^{94}} + \frac{1}{x} - 1$

Put $x = -1$

$$= \frac{1}{(-1)^{99}} + \frac{1}{(-1)^{98}} + \frac{1}{(-1)^{97}} + \frac{1}{(-1)^{96}}$$

$$+ \frac{1}{(-1)^{95}} + \frac{1}{(-1)^{94}} + \frac{1}{(-1)} - 1$$

$$= -1 + 1 - 1 + 1 - 1 + 1 - 1 - 1 = -2$$

14. (c); Given, $x = 6$

$$\frac{4 \times 6}{3} + 2P = 12$$

$$\Rightarrow 8 + 2P = 12$$

$$2P = 4, \quad P = 2$$

15. (b); $a^{1/3} = 11 \Rightarrow a = 11^3 = 1331$

$$\therefore a^2 - 331a = a(a - 331)$$

$$= 1331(1331 - 331)$$

$$= 1331 \times 1000 = 1331000$$

16. (b); $x + \frac{1}{x} = 2$

$x = 1$, satisfies the above equation

$$\therefore \text{put } x = 1 \text{ in } x^2 + \frac{1}{x^3} = (1)^2 + \frac{1}{(1)^3} = 2$$

17. (b); $x^2 - 3x + 1 = 0$

$$x^2 + 1 = 3x$$

Divide whole equation by x

$$x + \frac{1}{x} = 3, \quad x^3 + \frac{1}{x^3} = (3)^3 - 3(3) = 27 - 9 = 18$$

18. (c); $m + \frac{1}{m-2} = 4$

$$(m-2) + \frac{1}{(m-2)} = 4 - 2, \quad (m-2) + \frac{1}{(m-2)} = 2$$

Here, $(m-2) = 1$

$$\Rightarrow (m-2)^2 + \frac{1}{(m-2)^2} = 1^2 + \frac{1}{1^2} = 2$$

19. (b); $a + \frac{1}{a} = \sqrt{3}$

$$a^3 + \frac{1}{a^3} = (\sqrt{3})^3 - 3\sqrt{3} = 3\sqrt{3} - 3\sqrt{3} = 0$$

20. (b); $\frac{a}{b} + \frac{b}{a} - 1 = 0$

$$a^2 + b^2 - ab = 0$$

$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

$$= (a+b) \times 0$$

$$a^3 + b^3 = 0$$

21. (c); Here $x = 3, y = 5, z = 4$

$$\frac{x^2}{9} + \frac{y^2}{25} + \frac{z^2}{16} = \frac{9}{9} + \frac{25}{25} + \frac{16}{16} = 1 + 1 + 1 = 3$$

22. (a); $\frac{a}{b} = \frac{2}{3}$ and $\frac{b}{c} = \frac{4}{5}$

$$\frac{a}{b} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}, \quad \frac{b}{c} = \frac{4 \times 3}{5 \times 3} = \frac{12}{15}$$

Required ratio,

$$\frac{a+b}{b+c} = \frac{8+12}{12+15} = \frac{20}{27}$$

23. (a); $\frac{y}{x} = \frac{4}{15}, \frac{x}{y} = \frac{15}{4}$

By componendo and dividendo

$$\frac{x-y}{x+y} = \frac{15-4}{15+4} = \frac{11}{19}$$

24. (b); Given $\frac{3x+2y}{3x-2y} = \frac{4}{3}$

$$\Rightarrow 9x + 6y = 12x - 8y$$

$$14y = 3x$$

$$= \frac{\frac{x^2}{y^2} + 1}{\frac{x^2}{y^2} - 1} = \frac{\left(\frac{14}{3}\right)^2 + 1}{\left(\frac{14}{3}\right)^2 - 1} = \frac{205}{187}$$

25. (d); Given, $x + \frac{1}{x} = 5$

$$\Rightarrow \frac{3x}{2x^2 + 2 - 5x} = \frac{3x}{x \left[\left(2x + \frac{2}{x}\right) - 5 \right]}$$

$$\Rightarrow \frac{3}{2 \left(x + \frac{1}{x} \right) - 5} = \frac{3}{10 - 5} = \frac{3}{5}$$

26. (c); $\left(a + \frac{1}{a}\right)^2 = 3$

$$a + \frac{1}{a} = \sqrt{3}, \quad a^3 + \frac{1}{a^3} = 3\sqrt{3} - 3\sqrt{3}$$

$$a^3 + \frac{1}{a^3} = 0$$

$$\therefore a^3 + \frac{1}{a^3} + 3\sqrt{3} = 0 + 3\sqrt{3} = 3\sqrt{3}$$

27. (a); $P = \frac{x^2 - 36}{x^2 - 49}$

$$= \frac{(x+6)(x-6)}{(x+7)(x-7)}$$

$$Q = \frac{x+6}{x+7}, \quad P = \frac{Q \times (x-6)}{(x-7)}, \quad \frac{P}{Q} = \frac{x-6}{x-7}$$

28. (d); If $x + \frac{1}{x} = 2$

then $x = 1$

$$x^{17} + \frac{1}{x^{19}} = (1)^{17} + \frac{1}{(1)^{19}} = 1 + 1 = 2$$

29. (b); Given $\frac{2x^4 - 162}{(x^2 + 9)(2x - 6)}$

$$= \frac{2(x^4 - 81)}{2(x^2 + 9)(x - 3)} = \frac{2(x^2 - 9)(x^2 + 9)}{2(x^2 + 9)(x - 3)}$$

$$= \frac{x^2 - 9}{x - 3} = \frac{(x+3)(x-3)}{x-3} = x + 3$$

$$30. \text{ (b); Given, } \frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2ab - c^2} = \frac{a^2 - (b+c)^2}{(a+b)^2 - c^2}$$

$$= \frac{[a + (b+c)][a - (b+c)]}{(a+b+c)(a+b-c)} = \frac{a-b-c}{a+b-c}$$

31. (a); Given,

$$\frac{x + \frac{1}{x}}{2} = 1, \quad x + \frac{1}{x} = 2$$

then $x = 1$,

$$\therefore 8(1)^{10} + \frac{4}{(1)^5} = 8 + 4 = 12$$

32. (a); $x + y + z = 0$

$$\Rightarrow x + y = -z, y + z = -x, x + z = -y$$

$$\therefore (x+y)(y+z)(z+x) = (-z)(-x)(-y) = -xyz$$

33. (c); $x^2 - 4x + 1 = 0$

$$x^2 + 1 = 4x$$

Dividing the equation by x

$$x + \frac{1}{x} = 4$$

$$x^3 + \frac{1}{x^3} = a^3 - 3a$$

$$= 4^3 - 3(4) = 64 - 12 = 52$$

34. (a); Given $x + y + z = 6$ and $xy + yz + zx = 11$

$$x^3 + y^3 + z^3 - 3xyz$$

$$= (x+y+z)(x^2+y^2+z^2 - xy - yz - zx)$$

$$= (x+y+z)[(x+y+z)^2 - 3(xy+yz+zx)]$$

$$= 6[6^2 - 3(11)] = 6 \times 3 = 18$$

35. (a); $a^3 + b^3 + c^3 - 3abc$

$$= (a+b+c)(a^2+b^2+c^2 - ab - bc - ca)$$

$$= \frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= \frac{1}{2}(258 + 260 + 262)[(-2)^2 + (-2)^2 + 4^2]$$

$$= \frac{1}{2} \times 780 \times 24 = 9360$$

36. (b); Given, $pq + qr + rp = 0$

$$\Rightarrow -qr = pq + rp$$

$$\therefore \frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - rp} + \frac{r^2}{r^2 - pq}$$

$$= \frac{p^2}{p^2 + rp + pq} + \frac{q^2}{q^2 + pq + qr} + \frac{r^2}{r^2 + qr + rp}$$

$$= \frac{p}{p+q+r} + \frac{q}{p+q+r} + \frac{r}{p+q+r} = \frac{p+q+r}{p+q+r} = 1$$

37. (c); Given,

$$u^3 + (-2v)^3 + (-3w)^3 = 3 \times (-2)(-3)uvw$$

$$\therefore u + (-2v) + (-3w) = 0$$

$$u - 2v - 3w = 0$$

$$u - 2v = 3w$$

38. (c); $(x^2 + y^2)(p^2 + q^2) = (xp + yp)^2$

$$x^2p^2 + x^2q^2 + y^2p^2 + y^2q^2 = x^2p^2 + y^2p^2 + 2xpyq$$

$$x^2q^2 + y^2p^2 - 2xpyq = 0$$

$$(xq - yp)^2 = 0$$

$$xq = yp$$

39. (b); $(ab + bc + ca)^2$

$$= a^2b^2 + b^2c^2 + c^2a^2 + 2abbc + 2abca + 2bcca$$

$$\text{Or, } 0 = a^2b^2 + b^2c^2 + c^2a^2 + 2abc(b+a+c)$$

$$0 = a^2b^2 + b^2c^2 + c^2a^2 + 2pq$$

$$= a^2b^2 + b^2c^2 + c^2a^2 = -2pq$$

40. (b); $\frac{x}{y} = \frac{(p+q)^2}{(p-q)^2}$

By componendo and dividendo

$$\frac{x-y}{x+y} = \frac{(p+q)^2 - (p-q)^2}{(p+q)^2 + (p-q)^2} = \frac{4pq}{2(p^2 + q^2)}$$

$$= \frac{2pq}{p^2 + q^2}$$

Moderate

1. (a); Here, $\frac{a}{b} + \frac{b}{a} = 1$

$$\frac{a^2 + b^2}{ab} = 1, \quad a^2 + b^2 = ab$$

$$a^2 + b^2 - ab = 0$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$a^3 + b^3 = (a + b) \times 0$$

$$a^3 + b^3 = 0, \quad a^3 + b^3 + 2 = 2$$

2. (a); $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 4$

Here $x = y = 1$, satisfies the above equation
then $x^2 + y^2 = 1^2 + 1^2 = 2$

3. (b); $x^2 = y + z$

$$\Rightarrow x^2 + x = x + y + z$$

$$\Rightarrow x(x + 1) = x + y + z \quad \dots(i)$$

Similarly,

$$y(y + 1) = x + y + z \quad \dots(ii)$$

and

$$z(z + 1) = x + y + z \quad \dots(iii)$$

On adding (i), (ii), and (iii)

$$\therefore \frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$$

$$= \frac{x}{x+y+z} + \frac{y}{x+y+z} + \frac{z}{x+y+z} = \frac{x+y+z}{x+y+z} = 1$$

4. (a); $x + \frac{1}{x} = \sqrt{3}$

Cubing both sides

$$x^3 + \frac{1}{x^3} = 3\sqrt{3} - 3\sqrt{3}$$

$$x^6 + 1 = 0$$

$$\text{Now, } x^{18} + x^{12} + x^6 + 1$$

$$= x^{12}(x^6 + 1) + (x^6 + 1) = x^{12} \times 0 + 0 = 0$$

5. (d); $(ad - bc)^2 + (ac + bd)^2$

$$= a^2d^2 + b^2c^2 - 2acbd + a^2c^2 + b^2d^2 + 2abcd$$

$$= a^2d^2 + b^2c^2 + a^2c^2 + b^2d^2$$

$$= d^2(a^2 + b^2) + c^2(a^2 + b^2)$$

$$= (c^2 + d^2)(a^2 + b^2) = 1 \times 2 = 2$$

6. (a); $x + \frac{2}{3 + \frac{4}{5 + \frac{7}{6}}} = 10$

$$x + \frac{2}{3 + \frac{4 \times 6}{30 + 7}} = 10, \quad x + \frac{2}{3 + \frac{24}{37}} = 10$$

$$x + \frac{2 \times 37}{111 + 24} = 10, \quad x + \frac{2 \times 37}{135} = 10$$

$$x + \frac{74}{135} = 10, \quad x = 10 - \frac{74}{135}$$

$$x = \frac{1350 - 74}{135} = \frac{1276}{135}$$

7. (d); $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = 3$

$$\Rightarrow a = 3b, \quad c = 3d, \quad e = 3f$$

$$\therefore \frac{2a^2 + 3c^2 + 4e^2}{2b^2 + 3d^2 + 4f^2}$$

$$= \frac{2 \times 9b^2 + 3 \times 9d^2 + 4 \times 9f^2}{2b^2 + 3d^2 + 4f^2}$$

$$= \frac{9(2b^2 + 3d^2 + 4f^2)}{2b^2 + 3d^2 + 4f^2} = 9$$

8. (a); Given $a + \frac{1}{a} = 6$

$$\therefore a^4 + \frac{1}{a^4} = \left(a^2 + \frac{1}{a^2}\right)^2 - 2$$

$$= \left[\left(a + \frac{1}{a}\right)^2 - 2\right]^2 - 2$$

$$= [6^2 - 2]^2 - 2 = (34)^2 - 2 = 1154$$

9. (d); $x^4 + \frac{1}{x^4} = 23$

$$\text{Or, } \left(x^2 + \frac{1}{x^2}\right)^2 = 23 + 2$$

$$x^2 + \frac{1}{x^2} = \sqrt{25} = 5$$

$$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2, = 5 - 2$$

$$\left(x - \frac{1}{x}\right)^2 = 3$$

10. (c); $x^3 + y^3 = 35$ and $x + y = 5$
 $\Rightarrow (x + y)(x^2 + y^2 - xy) = 35$

$x^2 + y^2 - xy = 7$... (i)

and, $(x + y)^2 = 25$
 $x^2 + y^2 + 2xy = 25$... (ii)

On solving equation (i) and (ii)

$3xy = 18$

$xy = 6$... (iii)

$\Rightarrow x + y = 5$... (iv)

from (iii) and (iv)

$\frac{x+y}{xy} = \frac{5}{6}, \quad \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$

11. (c); $x^2 + y^2 + z^2 = 2x - 2y - 2z - 3$
 $x^2 + y^2 + z^2 - 2x + 2y + 2z + 1 + 1 + 1 = 0$
 $(x^2 - 2x + 1) + (y^2 + 2y + 1) + (z^2 + 2z + 1) = 0$
 $(x - 1)^2 + (y + 1)^2 + (z + 1)^2 = 0$
 $\therefore x = 1, y = -1$ and $z = -1$
 $5(1) - 4(-1) + 2(-1)$
 $5 + 4 - 2, 9 - 2 = 7$

12. (b); $2p + \frac{1}{p} = 4$

$\Rightarrow p + \frac{1}{2p} = 2$

$\left(p + \frac{1}{2p}\right)^3 = p^3 + \frac{1}{8p^3} + 3(p) \frac{1}{2p} \left(p + \frac{1}{2p}\right)$

$2^3 = p^3 + \frac{1}{8p^3} + \frac{3}{2} \times (2), \quad p^3 + \frac{1}{8p^3} = 8 - 3 = 5$

13. (b); $x = 3 + \sqrt{8}$ and

$\frac{1}{x} = \frac{1}{3 + \sqrt{8}} \times \frac{3 - \sqrt{8}}{3 - \sqrt{8}} = \frac{3 - \sqrt{8}}{9 - 8} = 3 - \sqrt{8}$

$x + \frac{1}{x} = 3 + \sqrt{8} + 3 - \sqrt{8}, \quad x + \frac{1}{x} = 6$

$x^2 + \frac{1}{x^2} = 6^2 - 2 = 36 - 2 = 34$

14. (c); Given

$\frac{2a+b}{a+4b} = 3$

$2a + b = 3a + 12b$

$a = -11b$

Then,

$\frac{a+b}{a+2b} = \frac{-11b+b}{-11b+2b} = \frac{-10b}{-9b} = \frac{10}{9}$

15. (c); Here, $a = 4.965, b = 2.343, c = 2.622$

$a - b - c = 4.965 - 2.343 - 2.622$
 $= 4.965 - 4.965$

$a - b - c = 0$

When $a - b - c = 0$ then

$a^3 - b^3 - c^3 - 3abc = 0$

16. (b); $x * y = 3x + 2y,$
 $2 * 3 + 3 * 4 = 3(2) + 2(3) + 3(3) + 2(4)$
 $= 6 + 6 + 9 + 8 = 29$

17. (b); $x^4 - 17x^3 + 17x^2 - 17x + 17$
 $= 16x^3 - 16x^3 - x^3 + 16x^2 + x^2 - 16x - x + 17$
 $= 16x^3 - 16x^3 - 16x^2 + 16x^2 + 16x - 16x - 16 + 17$
 $= 17 - 16 = 1$

18. (d); For maximum value

$a = b = c = d = \frac{1}{4}$

$(1 + a)(1 + b)(1 + c)(1 + d)$

$= \left(1 + \frac{1}{4}\right) \left(1 + \frac{1}{4}\right) \left(1 + \frac{1}{4}\right) \left(1 + \frac{1}{4}\right) = \left(\frac{5}{4}\right)^4$

19. (a); $\frac{1}{\sqrt[3]{4} + \sqrt[3]{2} + 1} = a\sqrt[3]{4} + b\sqrt[3]{2} + c$

$\Rightarrow \frac{1}{2^{\frac{2}{3}} + 2^{\frac{1}{3}} + 1} = a(2)^{\frac{2}{3}} + b(2)^{\frac{1}{3}} + c$

On multiplying numerator and denominator by

$(2^{\frac{1}{3}} - 1)$

$\Rightarrow \frac{(2^{\frac{1}{3}} - 1)}{(2^{\frac{1}{3}} - 1)(2^{\frac{2}{3}} + 2^{\frac{1}{3}} + 1)} = a \cdot 2^{\frac{2}{3}} + b \cdot 2^{\frac{1}{3}} + c$

$\frac{2^{\frac{1}{3}} - 1}{2 - 1} = a \cdot 2^{\frac{2}{3}} + b \cdot 2^{\frac{1}{3}} + c$

$\therefore a = 0, b = 1, c = -1$

$a + b + c = 0 + 1 - 1 = 0$

20. (a); $a^2 - b^2 + b^2 - c^2 + c^2 - a^2 = 0$

$\Rightarrow (a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$

$= 3(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$

$= 3(a - b)(a + b)(b - c)(b + c)(c - a)(c + a)$

$(a + b)(a - b)$ is the factor of given equation

21. (d); $p(p^2 + 3p + 3)$

$= p^3 + 3p^2 + 3p + 1 - 1$

$= (p + 1)^3 - 1 = (99 + 1)^3 - 1$

$= 100^3 - 1 = 1000000 - 1$

$= 999999$

$$\begin{aligned}
 22. \text{ (d); The expression is } & \sqrt[3]{p(p^2 + 3p + 3) + 1} \\
 &= \sqrt[3]{p^3 + 3p^2 + 3p + 1} = \left[(p+1)^3\right]^{\frac{1}{3}} = p+1 \\
 &= 123 + 1 = 124
 \end{aligned}$$

$$\begin{aligned}
 23. \text{ (d); } (a-b)^2 &= a^2 - 2ab + b^2 \\
 x^4 - 4x^2 + k &= (x^2)^2 - 2(x^2) \times 2 + 4 \\
 k &= 4
 \end{aligned}$$

$$24. \text{ (d); } x = 7 - 4\sqrt{3}$$

$$\frac{1}{x} = \frac{1}{7 - 4\sqrt{3}} = \frac{1(7 + 4\sqrt{3})}{(7 + 4\sqrt{3})(7 - 4\sqrt{3})}$$

$$= \frac{7 + 4\sqrt{3}}{49 - 48} = 7 + 4\sqrt{3}$$

$$x + \frac{1}{x} = 7 - 4\sqrt{3} + 7 + 4\sqrt{3}, \quad x + \frac{1}{x} = 14$$

$$25. \text{ (a); } x = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{(\sqrt{3} + 1)^2}{3 - 1} = \frac{3 + 1 + 2\sqrt{3}}{2}$$

$$= \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$$

$$\text{Similarly, } y = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = 2 - \sqrt{3}$$

$$\begin{aligned}
 \therefore x^2 + y^2 &= (2 + \sqrt{3})^2 + (2 - \sqrt{3})^2 \\
 &= 2(2^2 + (\sqrt{3})^2) = 2(4 + 3) = 14
 \end{aligned}$$

$$26. \text{ (b); } a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1} = \frac{\sqrt{5} + 1}{\sqrt{5} - 1} \times \frac{\sqrt{5} + 1}{\sqrt{5} + 1}$$

$$= \frac{(\sqrt{5} + 1)^2}{5 - 1} = \frac{5 + 1 + 2\sqrt{5}}{4}$$

$$a = \frac{3 + \sqrt{5}}{2}$$

Similarly,

$$b = \frac{3 - \sqrt{5}}{2}$$

$$a + b = \frac{3 + \sqrt{5}}{2} + \frac{3 - \sqrt{5}}{2} = 3$$

$$\text{and } ab = \frac{\sqrt{5} + 1}{\sqrt{5} - 1} \times \frac{\sqrt{5} - 1}{\sqrt{5} + 1} = 1$$

$$\begin{aligned}
 \frac{a^2 + ab + b^2}{a^2 - ab + b^2} &= \frac{(a+b)^2 - ab}{(a+b)^2 - 3ab} \\
 &= \frac{3^2 - 1}{3^2 - 3} = \frac{9 - 1}{9 - 3} = \frac{8}{6} = \frac{4}{3}
 \end{aligned}$$

$$27. \text{ (b); Given } x - \frac{1}{x} = 4$$

$$\left(x + \frac{1}{x}\right)^2 = \left(x - \frac{1}{x}\right)^2 + 4 = 4^2 + 4 = 16 + 4$$

$$\left(x + \frac{1}{x}\right)^2 = 20$$

$$x + \frac{1}{x} = \sqrt{20} = \sqrt{2 \times 2 \times 5} = 2\sqrt{5}$$

$$28. \text{ (b); } x = 3 + 2\sqrt{2}$$

$$\frac{1}{x} = \frac{1}{3 + 2\sqrt{2}} = \frac{1}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$\frac{1}{x} = 3 - 2\sqrt{2}, \quad x + \frac{1}{x} = 3 + 2\sqrt{2} + 3 - 2\sqrt{2}$$

$$x + \frac{1}{x} = 6$$

$$\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} - 2, \quad \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = 6 - 2 = 4$$

$$\sqrt{x} - \frac{1}{\sqrt{x}} = 2$$

$$29. \text{ (c); } x + \frac{1}{2x} = 2$$

Multiplying the above equation by 2

$$2x + \frac{1}{x} = 4$$

On cubing both sides

$$\left(2x + \frac{1}{x}\right)^3 = (2x)^3 + \left(\frac{1}{x}\right)^3 + 3(2x)\left(\frac{1}{x}\right)\left(2x + \frac{1}{x}\right)$$

$$64 = 8x^3 + \frac{1}{x^3} + 6 \times 4$$

$$8x^3 + \frac{1}{x^3} = 64 - 24 = 40$$

$$30. (a); \left(x + \frac{1}{x}\right)^2 = 3, \quad x + \frac{1}{x} = \sqrt{3}$$

$$x^3 + \frac{1}{x^3} = (\sqrt{3})^3 - 3\sqrt{3}$$

$$x^3 + \frac{1}{x^3} = 3\sqrt{3} - 3\sqrt{3}$$

$$x^6 + 1 = 0$$

$$\begin{aligned} x^{72} + x^{66} + x^{54} + x^{48} + x^{36} + x^{30} + x^{24} + x^{18} + x^6 + 1 \\ = x^{66}(x^6 + 1) + x^{48}(x^6 + 1) + x^{30}(x^6 + 1) + x^{18}(x^6 + 1) \\ + x^6 + 1 \\ = x^{66} \times 0 + x^{48} \times 0 + x^{30} \times 0 + x^{18} \times 0 + 0 = 0 \end{aligned}$$

$$31. (b); n = 7 + 4\sqrt{3} = 7 + 2 \times 2 \times \sqrt{3}$$

$$n = 4 + 3 + 2 \times 2\sqrt{3} = (2 + \sqrt{3})^2, \quad \sqrt{n} = 2 + \sqrt{3}$$

$$\frac{1}{\sqrt{n}} = \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}, \quad \frac{1}{\sqrt{n}} = 2 - \sqrt{3}$$

$$\therefore \sqrt{n} + \frac{1}{\sqrt{n}} = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$$

$$32. (a); \frac{x}{b+c} = \frac{y}{c+a} = \frac{x-y}{b+c-c-a} = \frac{x-y}{b-a}$$

$$\frac{y}{c+a} = \frac{z}{b+a} = \frac{y-z}{c+a-b-a} = \frac{y-z}{c-b}$$

$$\frac{z}{b+a} = \frac{x}{b+c} = \frac{z-x}{b+a-b-c} = \frac{z-x}{a-c}$$

$$\therefore \frac{x-y}{b-a} = \frac{y-z}{c-b} = \frac{z-x}{a-c}$$

$$33. (b); x^2 + y^2 + z^2 = 2x - 2y - 2z - 3$$

$$x^2 + y^2 + z^2 - 2x + 2y + 2z + 1 + 1 + 1 = 0$$

$$(x^2 - 2x + 1) + (y^2 + 2y + 1) + (z^2 + 2z + 1) = 0$$

$$(x-1)^2 + (y+1)^2 + (z+1)^2 = 0$$

$$\therefore x=1, \quad y=-1 \quad \text{and} \quad z=-1$$

$$2x - 3y + 4z = 2 + 3 - 4 = 1$$

$$34. (c); x + y + z = 6$$

On Squaring,

$$(x + y + z)^2 = 36$$

$$x^2 + y^2 + z^2 + 2xy + 2yz + 2zx = 36$$

$$20 + 2(xy + yz + zx) = 36$$

$$2(xy + yz + zx) = 16$$

$$xy + yz + zx = 8$$

$$\therefore x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$= 6(20 - 8) = 6 \times 12 = 72$$

$$35. (c); x + \frac{1}{x} = 99$$

$$\therefore \frac{100x}{2x^2 + 102x + 2} = \frac{100x}{2x^2 + 2 + 102x}$$

On dividing by x,

$$= \frac{100}{2x + \frac{2}{x} + 102} = \frac{100}{2\left(x + \frac{1}{x}\right) + 102}$$

$$= \frac{100}{2 \times 99 + 102} = \frac{100}{300} = \frac{1}{3}$$

$$36. (b); x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$$

$$3x - 2 = \frac{3}{x} \Rightarrow 3x - \frac{3}{x} = 2$$

$$\Rightarrow x - \frac{1}{x} = \frac{2}{3}$$

On squaring both sides

$$\left(x - \frac{1}{x}\right)^2 = \frac{4}{9} \Rightarrow x^2 + \frac{1}{x^2} - 2 = \frac{4}{9}$$

$$x^2 + \frac{1}{x^2} = \frac{4}{9} + 2 = 2\frac{4}{9}$$

$$37. (a); x^2 - 3x + 1 = 0$$

$$x + \frac{1}{x} = 3$$

On squaring both sides

$$\left(x + \frac{1}{x}\right)^2 = 9$$

$$x^2 + \frac{1}{x^2} + 2 = 9, \quad x^2 + \frac{1}{x^2} = 7$$

$$\Rightarrow x^2 + x + \frac{1}{x} + \frac{1}{x^2}$$

$$= \left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right)$$

$$= 7 + 3 = 10$$

$$38. (d); x - a = \frac{1}{x - b}$$

$$\Rightarrow (x - a)^3 - \frac{1}{(x - a)^3}$$

$$\begin{aligned}
&= \left(x - a - \frac{1}{x-a}\right)^3 + 3\left(x - a - \frac{1}{x-a}\right) \\
&= (x - a - x + b)^3 + 3(x - a - x + b) \\
&= (b - a)^3 + 3(b - a) \\
&= 5^3 + 3 \times 5 = 125 + 15 = 140
\end{aligned}$$

39. (c); $\frac{x}{y} = \frac{3}{2}$

$$\frac{x^2}{y^2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}, \quad \frac{2x^2 + 3y^2}{3x^2 - 2y^2} = \frac{2\left(\frac{x^2}{y^2}\right) + 3}{3\left(\frac{x^2}{y^2}\right) - 2}$$

On dividing numerator and denominator by y^2 .
Now,

$$\begin{aligned}
&= \frac{\left(2 \times \frac{9}{4}\right) + 3}{\left(3 \times \frac{9}{4}\right) - 2} = \frac{\frac{9}{2} + 3}{\frac{27}{4} - 2} \\
&= \frac{\frac{9+6}{2}}{\frac{27-8}{4}} = \frac{15}{2} \times \frac{4}{19} = \frac{30}{19} = 30:19
\end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{9+6}{2}}{\frac{27-8}{4}} = \frac{15}{2} \times \frac{4}{19} = \frac{30}{19} = 30:19
\end{aligned}$$

40. (d); Given, D : C = 6 : 5

C : B = 4 : 3 and B : A = 2 : 1

$\therefore$ D : C : B : A

$$\begin{aligned}
&= 6 \times 4 \times 2 : 5 \times 4 \times 2 : 5 \times 3 \times 2 : 5 \times 3 \times 1 \\
&= 48 : 40 : 30 : 15
\end{aligned}$$

Difficult

1. (d); $\frac{\sqrt{x+4} + \sqrt{x-4}}{\sqrt{x+4} - \sqrt{x-4}} = 2$

By componendo and dividendo,

$$\frac{2\sqrt{x+4}}{2\sqrt{x-4}} = \frac{2+1}{2-1}, \quad \frac{\sqrt{x+4}}{\sqrt{x-4}} = \frac{3}{1}$$

On squaring both sides,

$$\frac{x+4}{x-4} = \frac{9}{1}$$

$$x+4 = 9x-36$$

$$9x-x = 36+4$$

$$\Rightarrow 8x = 40$$

$$\Rightarrow x = 5$$

2. (c); $x = \sqrt[3]{a + \sqrt{a^2 + b^3}} + \sqrt[3]{a - \sqrt{a^2 + b^3}}$

On cubing both sides,

$$\begin{aligned}
x^3 &= \left(\sqrt[3]{a + \sqrt{a^2 + b^3}} + \sqrt[3]{a - \sqrt{a^2 + b^3}}\right)^3 \\
&= \left(\sqrt[3]{a + \sqrt{a^2 + b^3}}\right)^3 + \left(\sqrt[3]{a - \sqrt{a^2 + b^3}}\right)^3 +
\end{aligned}$$

$$3\left(\sqrt[3]{a + \sqrt{a^2 + b^3}}\right)\left(\sqrt[3]{a - \sqrt{a^2 + b^3}}\right)$$

$$\left[\sqrt[3]{a + \sqrt{a^2 + b^3}} + \sqrt[3]{a - \sqrt{a^2 + b^3}}\right]$$

$$= a + \sqrt{a^2 + b^3} + a - \sqrt{a^2 + b^3} +$$

$$3\left[\left(a + \sqrt{a^2 + b^3}\right)\left(a - \sqrt{a^2 + b^3}\right)\right]^{\frac{1}{3}} x$$

$$= 2a + 3(a^2 - a^2 - b^3)^{\frac{1}{3}} x$$

$$\therefore x^3 = 2a + 3(-b)x$$

$$x^3 + 3bx = 2a$$

3. (d); L.H.S = $\frac{4\sqrt{3} + 5\sqrt{2}}{\sqrt{48} + \sqrt{18}} = \frac{4\sqrt{3} + 5\sqrt{2}}{4\sqrt{3} + 3\sqrt{2}}$

On Rationalization

$$= \frac{4\sqrt{3} + 5\sqrt{2}}{4\sqrt{3} + 3\sqrt{2}} \times \frac{4\sqrt{3} - 3\sqrt{2}}{4\sqrt{3} - 3\sqrt{2}}$$

$$= \frac{48 - 12\sqrt{6} + 20\sqrt{6} - 30}{48 - 18}$$

$$= \frac{18 + 8\sqrt{6}}{30} = \frac{9}{15} + \frac{4\sqrt{6}}{15}$$

$$= \frac{3}{5} + \frac{4\sqrt{6}}{15}$$

$$\therefore \frac{3}{5} + \frac{4\sqrt{6}}{15} = a + b\sqrt{6}$$

$$\Rightarrow a = \frac{3}{5} \text{ and } b = \frac{4}{15}$$

4. (c); $a + \frac{1}{b} = 1 \Rightarrow a = 1 - \frac{1}{b} = \frac{b-1}{b}$

$$\Rightarrow \frac{1}{a} = \frac{b}{b-1} \text{ and}$$

$$b + \frac{1}{c} = 1 \Rightarrow \frac{1}{c} = 1 - b$$

$$c = \frac{1}{1-b}$$

$$\begin{aligned} \therefore c + \frac{1}{a} &= \frac{1}{1-b} + \frac{b}{b-1} \\ &= \frac{1}{1-b} - \frac{b}{1-b} = \frac{1-b}{1-b} = 1 \end{aligned}$$

5. (c); $a + b + c = 0 \Rightarrow b + c = -a$

On squaring both sides

$$\begin{aligned} \Rightarrow (b+c)^2 &= a^2 \\ b^2 + c^2 + 2bc &= a^2 \\ a^2 + b^2 + c^2 + 2bc &= 2a^2 \\ a^2 + b^2 + c^2 &= 2a^2 - 2bc = 2(a^2 - bc) \\ \frac{a^2 + b^2 + c^2}{a^2 - bc} &= \frac{2(a^2 - bc)}{a^2 - bc} = 2 \end{aligned}$$

6. (c); $x + \frac{1}{x} = 3$

On squaring both sides

$$\left(x + \frac{1}{x}\right)^2 = 9, \quad x^2 + \frac{1}{x^2} + 2 = 9$$

$$x^2 + \frac{1}{x^2} = 7 \quad \dots (i)$$

$$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$$

$$27 = x^3 + \frac{1}{x^3} + 3(3), \quad x^3 + \frac{1}{x^3} = 27 - 9$$

$$x^3 + \frac{1}{x^3} = 18 \quad \dots (ii)$$

On multiplying (i) and (ii)

$$\left(x^2 + \frac{1}{x^2}\right)\left(x^3 + \frac{1}{x^3}\right) = 7 \times 18 = 126$$

$$x^5 + \frac{1}{x^5} + x + \frac{1}{x} = 126$$

$$x^5 + \frac{1}{x^5} = 126 - 3$$

$$x^5 + \frac{1}{x^5} = 123$$

7. (c); $\frac{xy}{x+y} = a \Rightarrow \frac{x+y}{xy} = \frac{1}{a}$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{a} \quad \dots (i)$$

$$\frac{xz}{x+z} = b \Rightarrow \frac{x+z}{xz} = \frac{1}{b}$$

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{b} \quad \dots (ii)$$

Similarly,

$$\frac{1}{z} + \frac{1}{y} = \frac{1}{c} \quad \dots (iii)$$

From (i) and (iii)

$$\frac{1}{a} - \frac{1}{x} = \frac{1}{c} - \frac{1}{z}$$

From equation (ii)

$$\frac{1}{a} - \frac{1}{x} = \frac{1}{c} - \frac{1}{b} + \frac{1}{x}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{x} = \frac{1}{a} - \frac{1}{c} + \frac{1}{b}$$

$$\frac{2}{x} = \frac{bc - ab + ac}{abc}$$

$$\Rightarrow x = \frac{2abc}{bc - ab + ac}$$

8. (d); $a : b = 4 : 5$ and $b : c = 15 : 16$

$$a : b : c = 4 \times 3 : 15 : 16$$

$$a : b : c = 12 : 15 : 16$$

$$\frac{c}{a} = \frac{16}{12} = \frac{4}{3}$$

$$\therefore \frac{18\left(\frac{c^2}{a^2}\right) - 7}{45\left(\frac{c^2}{a^2}\right) + 20} = \frac{18\left(\frac{4}{3}\right)^2 - 7}{45\left(\frac{4}{3}\right)^2 + 20}$$

$$= \frac{32 - 7}{80 + 20} = \frac{25}{100} = \frac{1}{4}$$

9. (d); $x^a \times x^b \times x^c = 1$

$$x^{a+b+c} = 1, \quad x^{a+b+c} = x^0$$

$$\Rightarrow a + b + c = 0$$

$$\text{Now, } a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^3 + b^3 + c^3 - 3abc = 0$$

$$a^3 + b^3 + c^3 = 3abc$$

10. (c); Given, $x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}}$

$$x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \sqrt{\frac{(\sqrt{5}+1)^2}{5-1}} = \frac{\sqrt{5}+1}{2}$$

$$\therefore 5x^2 - 5x - 1 = 5\left(\frac{\sqrt{5}+1}{2}\right)^2 - 5\left(\frac{\sqrt{5}+1}{2}\right) - 1$$

$$= 5\left(\frac{5+1+2\sqrt{5}}{4}\right) - \frac{5\sqrt{5}+5}{2} - 1$$

$$= 5\left(\frac{3+\sqrt{5}}{2}\right) - \frac{5\sqrt{5}+5}{2} - 1$$

$$= \frac{15+5\sqrt{5}-5\sqrt{5}-5-2}{2} = \frac{8}{2} = 4$$

11. (a); Given α and β are the roots of the equation

$$x^2 + px + q = 0$$

then summation of the roots

$$\alpha + \beta = \frac{-p}{1}$$

product of the roots

$$\alpha \times \beta = \frac{q}{1}$$

$$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{(-p)^2 - 2q}{q}$$

$$= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{p^2 - 2q}{q}$$

12. (c); Given $xy(x+y) = 1$

$$\Rightarrow x + y = \frac{1}{xy}$$

$$\text{So, } (x+y)^3 = \frac{1}{x^3y^3}$$

$$\frac{1}{x^3y^3} - x^3 - y^3 = (x+y)^3 - x^3 - y^3$$

$$= x^3 + y^3 + 3xy(x+y) - x^3 - y^3$$

$$= 3xy(x+y)$$

$$= 3xy \times \frac{1}{xy} = 3$$

13. (d); Given,

$$ax + by = 6 \quad \dots(i)$$

$$bx - ay = 2 \quad \dots(ii)$$

On squaring and adding both the equation

$$(ax + by)^2 + (bx - ay)^2 = 6^2 + 2^2$$

$$a^2x^2 + b^2y^2 + 2abxy + b^2x^2 + a^2y^2 - 2abxy = 40$$

$$x^2(a^2 + b^2) + y^2(a^2 + b^2) = 40$$

$$(x^2 + y^2)(a^2 + b^2) = 40$$

$$a^2 + b^2 = \frac{40}{4}, \quad a^2 + b^2 = 10$$

14. (d); $2a - \frac{2}{a} + 3 = 0, \quad 2a - \frac{2}{a} = -3, \quad a - \frac{1}{a} = -\frac{3}{2}$

On cubing both the sides

$$\left(a - \frac{1}{a}\right)^3 = \left(-\frac{3}{2}\right)^3$$

$$a^3 - \frac{1}{a^3} - 3 \times a \times \frac{1}{a} \left(a - \frac{1}{a}\right) = -\frac{27}{8}$$

$$a^3 - \frac{1}{a^3} = -\frac{27}{8} + 3 \left(-\frac{3}{2}\right) = \frac{-27-36}{8}$$

$$a^3 - \frac{1}{a^3} + 2 = \frac{-27-36}{8} + 2 = \frac{-27-36+16}{8}$$

$$a^3 - \frac{1}{a^3} + 2 = \frac{-47}{8}$$

15. (d); Let $p(x) = 2x^3 + ax^2 + 11x + a + 3$

If $p(x)$ is exactly divisible by $(2x - 1)$, then

$(2x - 1)$ is a factor of $p(x)$

$$p\left(\frac{1}{2}\right) = 0 \quad [\text{because } 2x - 1 = 0 \Rightarrow x = \frac{1}{2}]$$

$$\Rightarrow 2 \times \left(\frac{1}{2}\right)^3 + a \times \left(\frac{1}{2}\right)^2 + 11\left(\frac{1}{2}\right) + a + 3 = 0$$

$$\frac{1}{4} + \frac{a}{4} + \frac{11}{2} + a + 3 = 0$$

$$\frac{1+a+22+4a+12}{4} = 0$$

$$5a + 35 = 0, \quad 5a = -35, \quad a = -7$$

16. (c); Let $q(x) = px^2 + 5x + r$

Since $(x - 2)$ and $\left(x - \frac{1}{2}\right)$ are factors of $q(x)$

therefore,

$$q(2) = 0 \text{ and } q\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow \begin{aligned} p(2^2) + 5(2) + r &= 0 \\ 4p + 10 + r &= 0 \end{aligned} \quad \dots (i)$$

$$\Rightarrow \begin{aligned} p\left(\frac{1}{2}\right)^2 + 5\left(\frac{1}{2}\right) + r &= 0 \\ \frac{p}{4} + \frac{5}{2} + r &= 0 \end{aligned} \quad \dots (ii)$$

On solving (i) and (ii)

$$\begin{aligned} 4p + r &= p + 4r \\ 3p &= 3r, p = r \end{aligned}$$

$$\frac{p}{r} = 1$$

17. (c); $a + b + c = 2s$

$$\begin{aligned} c &= 2s - a - b = (s - a) + (s - b) \\ \therefore (s - a)^3 + (s - b)^3 + 3(s - a)(s - b)c &= (s - a)^3 + (s - b)^3 + 3(s - a)(s - b)[(s - a) + (s - b)] \\ &= [(s - a) + (s - b)]^3 = c^3 \end{aligned}$$

18. (c); $x(x + y + z) = 28 \quad \dots (i)$

$$y(x + y + z) = 70 \quad \dots (ii)$$

$$z(x + y + z) = 98 \quad \dots (iii)$$

Adding all three equations

$$(x + y + z)(x + y + z) = 28 + 70 + 98$$

$$(x + y + z)^2 = 196$$

$$x + y + z = 14$$

$$\Rightarrow 3(x + y + z) = 3 \times 14 = 42$$

19. (d); $a^x = (x + y + z)^y, a^y = (x + y + z)^z, a^z = (x + y + z)^x$

Multiplying the above equations

$$a^x \times a^y \times a^z = (x + y + z)^{x+y+z}$$

$$a^{x+y+z} = (x + y + z)^{x+y+z}$$

$$a = x + y + z$$

20. (c); From first two ratio,

$$8(5x - 7y + 10) = 1(3x + 2y + 1)$$

$$\Rightarrow 37x - 58y = -79 \quad \dots (i)$$

From first and third ratio

$$9(5x - 7y + 10) = 1(11x + 4y - 10)$$

$$45x - 63y + 90 = 11x + 4y - 10$$

$$34x - 67y = -100 \quad \dots (ii)$$

from (i) and (ii)

$$\Rightarrow 3x + 9y = 21$$

$$x + 3y = 7 \quad \dots (iii)$$

putting, $x = 7 - 3y$ in equation (i)

$$37(7 - 3y) - 58y = -79$$

$$259 - 111y - 58y = -79$$

$$338 = 169y, y = 2$$

Again from (iii)

$$x + 3(2) = 7$$

$$\Rightarrow x = 7 - 6, x = 1$$

$$\therefore x + y = 1 + 2 = 3$$

21. (c); Average of x and $\frac{1}{x} = M$

$$\Rightarrow \frac{x + \frac{1}{x}}{2} = M, x + \frac{1}{x} = 2M$$

On squaring both sides

$$\left(x + \frac{1}{x}\right)^2 = (2M)^2, x^2 + \frac{1}{x^2} + 2 = 4M^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 4M^2 - 2$$

$$\text{Average of } x^2 + \frac{1}{x^2} = \frac{x^2 + \frac{1}{x^2}}{2}$$

$$= \frac{4M^2 - 2}{2} = \frac{2(2M^2 - 1)}{2} = 2M^2 - 1$$

22. (c); Let the each ratio be K

$$\frac{x}{(b - c)(b + c - 2a)} = \frac{y}{(c - a)(c + a - 2b)}$$

$$= \frac{z}{(a - b)(a + b - 2c)} = K$$

$$x = K(b - c)(b + c - 2a) = K(b^2 - c^2) - K2a(b - c)$$

$$y = K(c - a)(c + a - 2b) = K(c^2 - a^2) - K2b(c - a)$$

$$z = K(a - b)(a + b - 2c) = K(a^2 - b^2) - K2c(a - b)$$

On Adding, $x + y + z = 0$

23. (d); $x^8 + x^4 + 1 = x^8 + 2x^4 + 1 - x^4$

$$= (x^4 + 1)^2 - (x^2)^2$$

$$= (x^4 + 1 + x^2)(x^4 + 1 - x^2)$$

$$= (x^4 + 2x^2 + 1 - x^2)(x^4 - x^2 + 1)$$

$$= ((x^2 + 1)^2 - x^2)(x^4 - x^2 + 1)$$

$$= (x^2 + 1 - x)(x^2 + 1 + x)(x^4 - x^2 + 1)$$

$$= (x^2 - x + 1)(x^2 + 2x + 1 - x)(x^4 - x^2 + 1)$$

$$= (x^2 - x + 1)((x + 1)^2 - x)(x^4 - x^2 + 1)$$

$$= (x^2 - x + 1)(x + 1 + \sqrt{x})(x + 1 - \sqrt{x})(x^4 - x^2 + 1)$$

Among all options $x^2 - 2x + 1$ is not a factor.

24. (a); $pqr = 1$

$$\therefore r = p^{-1}q^{-1} \text{ and } r^{-1} = pq$$

Eliminating r from given expression,

$$\frac{1}{1 + p + q^{-1}} + \frac{1}{1 + q + r^{-1}} + \frac{1}{1 + r + p^{-1}}$$

$$= \frac{q}{q + pq + 1} + \frac{1}{1 + q + pq} + \frac{1}{1 + p^{-1}q^{-1} + p^{-1}}$$

$$= \frac{q}{1+q+pq} + \frac{1}{1+q+pq} + \frac{pq}{1+q+pq}$$

$$= \frac{1+q+pq}{1+q+pq} = 1$$

25. (c); $\left(\frac{y-z-x}{2}\right)^3 + \left(\frac{z-x-y}{2}\right)^3 + \left(\frac{x-y-z}{2}\right)^3$

$$= \left(\frac{y-(z+x)}{2}\right)^3 + \left(\frac{z-(x+y)}{2}\right)^3 + \left(\frac{x-(y+z)}{2}\right)^3$$

$$= \left(\frac{y-(-y)}{2}\right)^3 + \left(\frac{z-(-z)}{2}\right)^3 + \left(\frac{x-(-x)}{2}\right)^3$$

$$= \left(\frac{2y}{2}\right)^3 + \left(\frac{2z}{2}\right)^3 + \left(\frac{2x}{2}\right)^3$$

$$= y^3 + z^3 + x^3$$

(If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$)

$$x^3 + y^3 + z^3 = 3xyz$$

Previous Year (Memory Based)

1. (b); $x^4 + \frac{1}{x^4} = 23$

On adding 2 both sides

$$x^4 + \frac{1}{x^4} + 2 = 23 + 2$$

$$\left(x^2 + \frac{1}{x^2}\right)^2 = 25, \quad x^2 + \frac{1}{x^2} = 5$$

$$x^2 + \frac{1}{x^2} - 2 = 5 - 2, \quad \left(x - \frac{1}{x}\right)^2 = 3$$

2. (c); $(a+b)^2 + (a-b)^2 = 6^2 + 2^2$
 $2(a^2+b^2) = 40$

3. (c); $x^4 + \frac{1}{x^4} = 119$

On adding 2 both sides

$$x^4 + \frac{1}{x^4} + 2 = 119 + 2$$

$$\left(x^2 + \frac{1}{x^2}\right)^2 = 121, \quad x^2 + \frac{1}{x^2} = 11$$

$$x^2 + \frac{1}{x^2} - 2 = 11 - 2$$

$$\left(x - \frac{1}{x}\right)^2 = 9, \quad x - \frac{1}{x} = \pm 3$$

4. (d); $a + b + c = 8$

$$\Rightarrow (a-4) + (b-3) + (c-1) = 8 - 4 - 3 - 1 = 0$$

If $x + y + z = 0$ then

$$x^3 + y^3 + z^3 = 3xyz$$

Putting $x = a - 4$, $y = b - 3$ and $z = c - 1$

$$(a-4)^3 + (b-3)^3 + (c-1)^3 - 3(1-4)(b-3)$$

$$(c-1) = 0$$

5. (a); $x^4 + y^4 - 2x^2y^2 = (x^2 - y^2)^2$
 $= [(x-y)(x+y)]^2$

$$= \left(2\sqrt{a} \cdot \frac{2}{\sqrt{a}}\right)^2 = (4)^2 = 16$$

6. (d); $5a + \frac{1}{3a} = 5$

$$\Rightarrow \frac{5}{3} \left(3a + \frac{1}{5a}\right) = 5, \quad 3a + \frac{1}{5a} = 3$$

On squaring both sides

$$\left(3a + \frac{1}{5a}\right)^2 = 9, \quad 9a^2 + \frac{1}{25a^2} + \frac{6}{5} = 9$$

$$9a^2 + \frac{1}{25a^2} = 9 - \frac{6}{5} = \frac{45-6}{5} = \frac{39}{5}$$

7. (d); $\frac{x^4 + \frac{1}{x^2}}{x^2 - 3x + 1} = \frac{x \left(x^3 + \frac{1}{x^3}\right)}{x \left(x - 3 + \frac{1}{x}\right)} = \frac{x^3 + \frac{1}{x^3}}{x + \frac{1}{x} - 3}$

$$= \frac{\left(x + \frac{1}{x}\right)^3 - 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)}{\left(x + \frac{1}{x}\right) - 3}$$

$$= \frac{5^3 - 3 \cdot 1 \cdot 5}{5 - 3} = \frac{110}{2} = 55$$

8. (b); $= \frac{5x^2 - 3y^2}{xy} = \frac{11}{2}$

$$5\left(\frac{x}{y}\right) - 3\left(\frac{y}{x}\right) = \frac{11}{2}$$

If $\frac{x}{y} = a$ then $5a - \frac{3}{a} = \frac{11}{2}$

or $5a^2 - 3 = \frac{11}{2}a$

$10a^2 - 6 = 11a$, $10a^2 - 11a - 6 = 0$

$10a^2 - 15a + 4a - 6 = 0$, $5a(2a - 3) + 2(2a - 3) = 0$

$a = \frac{3}{2}, \frac{-2}{5}$

Since $\frac{x}{y}$ is positive

$\therefore \frac{x}{y} = \frac{3}{2}$

9. (b); $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$

$56 = 2(a^2 + b^2 + ab)$

$28 = a^2 + b^2 - 2ab + 3ab$

$28 = (a - b)^2 + 3ab$

$28 = 4 + 3ab$

$ab = \frac{24}{3} = 8$

$\therefore a^2 + b^2 = (a - b)^2 + 2ab$

$= 2^2 + 2 \times 8 = 4 + 16 = 20$

10. (b); $x = y \Rightarrow 2t = \frac{2t-1}{3}$

$6t = 2t - 1$

$\Rightarrow 6t = 2t - 1 \Rightarrow t = -\frac{1}{4}$

11. (a); $(x - y)^2$ is always greater than or equal to zero

$(x - y)^2 \geq 0$

$x^2 + y^2 - 2xy \geq 0$

$(x + y)^2 - 4xy \geq 0$

$8^2 - 4xy \geq 0$

$64 - 4xy \geq 0$

$xy \leq 16$

12. (d); $(2 + \sqrt{3})a = (2 - \sqrt{3})b = 1$

or $a = \frac{1}{2 + \sqrt{3}}, b = \frac{1}{2 - \sqrt{3}}$

$\frac{1}{a} = 2 + \sqrt{3}, \frac{1}{b} = 2 - \sqrt{3}$

$\frac{1}{a} + \frac{1}{b} = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$

13. (b); $3x + \frac{3}{x} = 1, x + \frac{1}{x} = \frac{1}{3}$

On cubing both sides

$x + \frac{1}{x} = \frac{1}{3}$

On Cubing the above equation

$\Rightarrow x^3 + \frac{1}{x^3} + 3 \times \frac{1}{3} = \frac{1}{27}$

$x^3 + \frac{1}{x^3} + 1 = \frac{1}{27}$

14. (b); We have, $\frac{3x}{2y} = \frac{21}{22}$

$\frac{x}{y} = \frac{7}{11}, x = \frac{7}{11}y$

Now, $4x + 5y = 83$

$4\left(\frac{7}{11}y\right) + 5y = 83$

$\frac{28}{11}y + 5y = 83$

$83y = 83 \times 11, y = 11$

Similarly, $x = \frac{7}{11} \times 11 = 7$

$\therefore y - x = 11 - 7 = 4$

15. (a); $a^2 + 4b^2 + 4b - 4ab - 2a - 8$

On adding and subtracting $2a$ and $4b$, we get

$= a^2 + 4b^2 + 4b - 4ab - 2a - 8 + 2a - 2a - 4b + 4b$

$= a^2 - 2ab + 2a - 2ab + 4b^2 - 4b - 4a + 8b - 8$

$= a(a - 2b + 2) - 2b(a - 2b + 2) - 4(a - 2b + 2)$

$= (a - 2b - 4)(a - 2b + 2)$

16. (c); Here, $p = \frac{5}{18}$

Consider, $27p^3 - \frac{1}{216} - \frac{9}{2}p^2 + \frac{1}{4}p$

$= (3p)^3 - \left(\frac{1}{6}\right)^3 - 3 \times (3p)^2 \times \frac{1}{6} + 3 \times 3p \times \frac{1}{36}$

$= \left(3p - \frac{1}{6}\right)^3 \quad [\because (a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2]$

$= \left(3 \times \frac{5}{18} - \frac{1}{6}\right)^3 = \left(\frac{4}{6}\right)^3 = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$

17. (d); Here, $x + \frac{1}{x} = 1$

$$\text{Then, } \frac{x^2 + 3x + 1}{x^2 + 7x + 1} = \frac{x \left(x + 3 + \frac{1}{x} \right)}{x \left(x + 7 + \frac{1}{x} \right)}$$

$$= \frac{x + \frac{1}{x} + 3}{x + \frac{1}{x} + 7} = \frac{1 + 3}{1 + 7} = \frac{4}{8} = \frac{1}{2}$$

18. (b); Given, $a = \sqrt{6} + \sqrt{5}$

$$b = \sqrt{6} - \sqrt{5}$$

$$a^2 = (\sqrt{6} + \sqrt{5})^2$$

$$= 6 + 5 + 2\sqrt{30} = 11 + 2\sqrt{30}$$

Similarly,

$$b^2 = (\sqrt{6} - \sqrt{5})^2$$

$$= 6 + 5 - 2\sqrt{30} = 11 - 2\sqrt{30}$$

$$\text{and, } ab = (\sqrt{6} + \sqrt{5})(\sqrt{6} - \sqrt{5}) = 6 - 5 = 1$$

$$\text{Therefore, } 2a^2 - 5ab + 2b^2$$

$$= 2(11 + 2\sqrt{30}) - 5(1) + 2(11 - 2\sqrt{30})$$

$$= 22 + 4\sqrt{30} - 5 + 22 - 4\sqrt{30} = 44 - 5 = 39$$

19. (a); $\sqrt{1 + \left(\frac{x^4 - 1}{-2x^2} \right)^2} = \sqrt{1 + \frac{(x^4 - 1)^2}{4x^4}}$

$$= \sqrt{\frac{4x^4 + x^8 + 1 - 2x^4}{4x^4}} = \sqrt{\left(\frac{x^4 + 1}{4x^4} \right)^2} = \frac{x^4 + 1}{2x^2}$$

20. (a); Here, $x^2 + 4x + 3 = 0$

$$x^2 + 3x + x + 3 = 0$$

$$x(x + 3) + 1(x + 3) = 0$$

$$(x + 1)(x + 3) = 0$$

$$\Rightarrow x = -1, -3$$

Now, $\frac{x^3}{x^6 + 27x^3 + 27}$

$$= \frac{x^3}{x^3 \left(x^3 + 27 + \frac{27}{x^3} \right)} = \frac{1}{x^3 + 27 + \frac{27}{x^3}}$$

Putting $x = -1$ in above equation.

$$= \frac{1}{(-1)^3 + 27 + \frac{27}{(-1)^3}}$$

$$= \frac{1}{-1 + 27 - 27} = \frac{1}{-1} = -1$$

21. (a); Given, $x + y = 4$... (i)

$$\text{and } \frac{1}{x} + \frac{1}{y} = 4$$

$$\Rightarrow \frac{y + x}{xy} = 4$$

$$\Rightarrow x + y = 4xy$$

$$\Rightarrow 4 = 4xy \quad [\text{from Eq. (i)}]$$

$$\Rightarrow xy = 1 \quad \dots (ii)$$

We know that,

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

$$x^3 + y^3 = (x + y)[(x^2 + y^2) - xy]$$

$$= (x + y)[(x + y)^2 - 2xy - xy]$$

$$= (4)[(4)^2 - 3 \times 1]$$

$$= 4(16 - 3) = 4 \times 13 = 52$$

22. (d); If $\frac{x}{xa + yb + zc} = \frac{y}{ya + zb + xc} = \frac{z}{za + xb + yc} = k$

$$\therefore x = k(xa + yb + zc)$$

$$y = k(ya + zb + xc)$$

$$\text{and } z = k(za + xb + yc)$$

Now,

$$x + y + z = k[a(x + y + z) + b(x + y + z) + c(x + y + z)]$$

$$\Rightarrow (x + y + z) = k[(x + y + z)(a + b + c)]$$

$$\Rightarrow (x + y + z) = k[(x + y + z)(a + b + c)]$$

$$\Rightarrow k = \frac{1}{a + b + c}$$

23. (d); Given, $x = 3 + 2\sqrt{2}$

$$\frac{1}{x} = \frac{1}{3 + 2\sqrt{2}} = \frac{1}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$= 3 - 2\sqrt{2} \quad (\text{on rationalising})$$

$$\therefore x^3 + \frac{1}{x^3} = (3 + 2\sqrt{2})^3 + (3 - 2\sqrt{2})^3$$

$$= 27 + 16\sqrt{2} + 3 \cdot 9 \cdot 2\sqrt{2} + 3 \cdot 3 \cdot 8 +$$

$$27 - 16\sqrt{2} - 3 \cdot 9 \cdot 2\sqrt{2} + 3 \cdot 3 \cdot 8$$

$$= 27 + 72 + 27 + 72 = 198$$

Now,

$$\begin{aligned}\therefore x^3 - \frac{1}{x^3} &= (3 + 2\sqrt{2})^3 - (3 - 2\sqrt{2})^3 \\ &= 27 + 16\sqrt{2} + 3 \cdot 9 \cdot 2\sqrt{2} + 3 \cdot 3 \cdot 8 - 27 + \\ &\quad 16\sqrt{2} + 3 \cdot 9 \cdot 2\sqrt{2} - 3 \cdot 3 \cdot 8 \\ &= 16\sqrt{2} + 54\sqrt{2} + 16\sqrt{2} + 54\sqrt{2} \\ &= 140\sqrt{2}\end{aligned}$$

24. (d); Taking expression $x + y = 3$

On cubing both sides, we get $(x + y)^3 = (3)^3$

$$\Rightarrow x^3 + y^3 + 3xy(x + y) = 27$$

$$\Rightarrow 9 + 3xy(3) = 27 \quad [\because x^3 + y^3 = 9 \text{ and } x + y = 3]$$

$$\Rightarrow 9xy = 18$$

$$\therefore xy = 2$$

Now, $(x + y) = 3$

On squaring both sides, we get

$$x^2 + y^2 + 2xy = 9$$

$$x^2 + y^2 + 2(2) = 9$$

$$x^2 + y^2 = 5 \quad [\because xy = 2]$$

Again, squaring on both sides, we get

$$(x^2 + y^2)^2 = (5)^2$$

$$\Rightarrow x^4 + y^4 + 2x^2y^2 = 25$$

$$\therefore x^4 + y^4 = 25 - 2(xy)^2$$

$$= 25 - 2(2)^2$$

$$= 25 - 8 = 17$$

25. (c); $x^4 - 2x^2y^2 + y^4 = [(x^2 - y^2)]^2$

$$= [(x + y)(x - y)]^2$$

$$= \left[\left(p + \frac{1}{p} + p - \frac{1}{p} \right) \left(p + \frac{1}{p} - p + \frac{1}{p} \right) \right]^2$$

[Putting the value of x and y]

$$= \left[(2p) \left(\frac{2}{p} \right) \right]^2 = 4p^2 \times \frac{4}{p^2} = 4 \times 4 = 16$$

26. (c); $x + \frac{1}{2x} = 2$

On multiplying by 2 both sides, we get

$$2x + \frac{2}{2x} = 4$$

$$\Rightarrow 2x + \frac{1}{x} = 4$$

On cubing both sides, we get

$$\left(2x + \frac{1}{x} \right)^3 = 4^3$$

$$\begin{aligned}\Rightarrow (2x)^3 + \left(\frac{1}{x} \right)^3 + 3 \times 2x \times \frac{1}{x} \left(2x + \frac{1}{x} \right) &= 64 \\ [\because (a + b)^3 &= a^3 + b^3 + 3ab(a + b)]\end{aligned}$$

$$8x^3 + \frac{1}{x^3} + 3 \times 2x \times \frac{1}{x} \left(2x + \frac{1}{x} \right) = 64$$

$$8x^3 + \frac{1}{x^3} + 6 \times 4 = 64$$

$$8x^3 + \frac{1}{x^3} = 64 - 24 = 40$$

27. (b); $\frac{x}{2x^2 + 5x + 2} = \frac{1}{6}$

$$\Rightarrow 2x^2 + 5x + 2 = 6x$$

$$\Rightarrow 2x^2 + 2 = 6x - 5x = x, \quad 2(x^2 + 1) = x$$

On dividing by 2, we get

$$x^2 + 1 = \frac{x}{2}$$

On dividing by x , we get

$$x + \frac{1}{x} = \frac{1}{2}$$

28. (a); $a + b + c = 0$

$$\Rightarrow b + c = -a$$

On squaring both sides, we get

$$(b + c)^2 = a^2$$

$$\Rightarrow b^2 + c^2 + 2bc = a^2$$

On adding a^2 both sides, we get

$$a^2 + b^2 + c^2 + 2bc = 2a^2$$

$$\Rightarrow a^2 + b^2 + c^2 = 2a^2 - 2bc$$

$$\Rightarrow a^2 + b^2 + c^2 = 2(a^2 - bc)$$

On dividing by $(a^2 - bc)$ both sides, we get

$$\frac{a^2 + b^2 + c^2}{a^2 - bc} = \frac{2(a^2 - bc)}{a^2 - bc} = 2$$

$$\frac{a^2 + b^2 + c^2}{a^2 - bc} - 2 = 2 - 2 = 0$$

29. (b); $5^{\sqrt{x}} + 12^{\sqrt{x}} = 13^{\sqrt{x}}$

We know that

$$5^2 + 12^2 = 13^2$$

$$\sqrt{x} = 2$$

$$x = 2^2 = 4$$

30. (a); Given, $x^2 + 2 = 2x$

$$\Rightarrow x^2 - 2x + 2 = 0$$

Now, Dividing $x^4 - x^3 + x^2 + 2$ by $x^2 - 2x + 2$

$$\begin{array}{r}
 x^2 - 2x + 2 \overline{) \begin{array}{l} x^4 - x^3 + x^2 + 2 \\ x^4 - 2x^3 + 2x^2 \end{array} } \begin{array}{l} x^2 + x + 1 \\ x^3 - x^2 + 2 \\ x^3 - 2x^2 + 2x \\ x^2 - 2x + 2 \\ x^2 - 2x + 2 \\ \hline x \end{array}
 \end{array}$$

Hence, $x^4 - x^3 + x^2 + 2 = (x^2 - 2x + 2)(x^2 + x + 1) = 0$

31. (b); $a : b = 2 : 3$, $b : c = 4 : 5$

or $a : b = 8 : 12$, $b : c = 12 : 15$

$\therefore a : b : c = 8 : 12 : 15$

$\therefore a^2 : b^2 : bc = 8^2 : 12^2 : 15 \times 12$

$64 : 144 : 180 = 16 : 36 : 45$

32. (b); Expression = $\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}}$

$$= \frac{(3^5)^{\frac{n}{5}} \times 3^{2n+1}}{(3^2)^n \times 3^{n-1}} = \frac{3^n \times 3^{2n+1}}{3^{2n} \times 3^{n-1}} = \frac{3^{n+2n+1}}{3^{2n+n-1}}$$

$$= \frac{3^{3n+1}}{3^{3n-1}} = 3^{3n+1-3n+1} = 3^2 = 9$$

33. (a); $x^2 - 3x + 1 = 0$

$$\Rightarrow x^2 + 1 = 3x$$

On dividing by x both sides, we get

$$x + \frac{1}{x} = 3$$

On cubing both sides, we get

$$\left(x + \frac{1}{x}\right)^3 = (3)^3$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3x \times \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$[\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)]$$

$$= (3)^3 - 3 \times 3 = 27 - 3 \times 3 = 18$$

$$x^3 + \frac{1}{x^3} - 9 = 18 - 9 = 9$$

34. (c); Given, $a^2 + b^2 + 2b + 4a + 5 = 0$

$$\Rightarrow a^2 + 4a + b^2 + 2b + 5 = 0$$

$$\Rightarrow a^2 + 4a + 4 + b^2 + 2b + 1 = 0$$

$$(a+2)^2 + (b+1)^2 = 0 \quad [\because (a+b)^2 = a^2 + b^2 + 2ab]$$

It is possible only, when

$$a + 2 = 0 \Rightarrow a = -2 \text{ and } b + 1 = 0$$

$$\Rightarrow b = -1$$

$$\therefore \frac{a-b}{a+b} = \frac{-2-1}{-2-1} = \frac{-1}{-3} = \frac{1}{3}$$

35. (c); Given, $x = 5 + 2\sqrt{6}$... (i)

$$\therefore \frac{1}{x} = \frac{1}{5+2\sqrt{6}} = \frac{5-2\sqrt{6}}{(5+2\sqrt{6})(5-2\sqrt{6})}$$

(rationalising)

$$= \frac{5-2\sqrt{6}}{25-24} = 5-2\sqrt{6} \quad \dots (ii)$$

$$[\because (a+b)(a-b) = a^2 - b^2]$$

$$\therefore \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = (\sqrt{x})^2 + \left(\frac{1}{\sqrt{x}}\right)^2 + 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}}$$

$$[\because (a+b)^2 = a^2 + b^2 + 2ab]$$

$$= x + \frac{1}{x} + 2 \quad [\text{from Eqs. (i) and (ii)}]$$

$$= 5 + 2\sqrt{6} + 5 - 2\sqrt{6} + 2 = 12$$

$$\therefore \sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{12} = 2\sqrt{3}$$

36. (d); $\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - 1\right)\left(x^2 + \frac{1}{x^2} + 1\right)$

$$= \left(x^2 - \frac{1}{x^2}\right)\left[\left(x^2 + \frac{1}{x^2}\right)^2 - 1\right]$$

$$[\because (a+b)(a-b) = a^2 - b^2]$$

$$= \left(x^2 - \frac{1}{x^2}\right)\left(x^4 + \frac{1}{x^4} + 1\right)$$

$$= x^6 + \frac{1}{x^2} + x^2 - x^2 - \frac{1}{x^6} - \frac{1}{x^2} = x^6 - \frac{1}{x^6}$$

37. (b); Given, $2p + \frac{1}{p} = 4$... (i)

On dividing by 2 both sides, we get

$$p + \frac{1}{2p} = 2 \quad \dots (ii)$$

On squaring both sides, we get

$$p^2 + \frac{1}{4p^2} + 2(p)\left(\frac{1}{2p}\right) = 4$$

$$p^2 + \frac{1}{4p^2} + 1 = 4, \quad p^2 + \frac{1}{4p^2} = 3$$

$$38. \text{ (c); } \because \frac{2a+b}{a+4b} = 3 \quad (\text{given})$$

$$\Rightarrow 2a + b = 3a + 12b$$

$$\Rightarrow 3a - 2a = b - 12b$$

$$\Rightarrow a = -11b$$

Then,

$$\frac{a-b}{a+2b} = \frac{-11b-b}{-11b+2b} = \frac{-12b}{-9b} = \frac{4}{3}$$

$$39. \text{ (c); } x^{\frac{1}{3}} + y^{\frac{1}{3}} - z^{\frac{1}{3}} = 0 \quad \dots(i)$$

$$x^{\frac{1}{3}} + y^{\frac{1}{3}} = z^{\frac{1}{3}}$$

On cubing both sides, we get

$$\left(x^{\frac{1}{3}} + y^{\frac{1}{3}}\right)^3 = z^{\frac{1}{3} \times 3}$$

$$\Rightarrow x + y + 3x^{\frac{1}{3}} \cdot y^{\frac{1}{3}} (x^{\frac{1}{3}} + y^{\frac{1}{3}}) = z$$

$$[\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)]$$

$$\Rightarrow x + y - z$$

$$= -3x^{\frac{1}{3}} \cdot y^{\frac{1}{3}} z^{\frac{1}{3}} \quad \dots(ii)$$

Now, Again taking cube both sides

$$(x + y - z)^3 = -27xyz$$

$$(x + y - z)^3 + 27xyz = 0$$

$$40. \text{ (c); Given, } 10^{0.48} = x \text{ and } 10^{0.70} = y$$

$$\therefore x^z = y^2$$

$$\Rightarrow (10^{0.48})^z = (10^{0.70})^2$$

$$10^{0.48z} = 10^{1.4}$$

$$0.48z = 1.4$$

$$z = \frac{140}{48} = 2.9$$